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AI as Co-author: towards balancing automation and designer agency

A case study at Liip SA

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Abstract

The rapid integration of artificial intelligence has greatly impacted the design industry and the relationship between service providers, employees and users. There is an increasing demand for greater efficiency but without a clear plan to implement AI-powered human-centred practices. The fear of AI replacement continues to grow amidst the design landscape and manifests in growing internal tension and underwhelming results. This case study was conducted on a boutique digital services company, specialised in web development and with increasing AI focused projects, with the goal of gathering empirical data of the use of AI in design. Central to this research is the exploration of how AI can be utilised as a "co-author" to improve internal processes while retaining designer agency. This research aims to bridge the gap between rapid technological adoption and task automation with the human essence that defines the high-quality of the design of services and products.

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1. Introduction

1.1 Motivation

When we hear the word “futuristic”, we imagine a world running on 100% efficiency. Cars drive by themselves, robots clean and cook, fridges make and order our shopping list that gets delivered by self-flying drones or a fully smart house where all the appliances are connected through an invisible web, and it is voice-activated! It is an easy world to live in, it does not need too much thinking, effort, or worrying, unless of course you have a dentist appointment. This is the satirical scenario posed by Rema 1000 in their marketing campaign “Smart House” (TRY, 2018). In the video, a man frantically struggles to get back into his voice-controlled, high-tech house after a dentist appointment affected his speaking. In the nearby house, his neighbour has no issue entering her house with a simple key.

It is hard to ignore the rapid wave of technological advancements from the past decade, and the desire to implement all of them in our daily lives is understandable. Artificial intelligence (AI) is being sold by many as a miracle drug to all our life's problems; it can detect hard-to-spot cancers, create deep relationships, and hyper-personalise our education (Reznik et al., 2026). It is here to stay, and it will continue to branch into every industry and sector of our lives. Of course, with all the advantages it brings us, AI currently poses a threat to our privacy and data security, our cognitive capabilities, our mental health, and our jobs. With more companies adopting this innovation in their products and services, we start to see the ugly side of using a technology with no strategy or plan on how to implement it. From AI chatbots and digital call centres that make it harder for users to get good customer service, to employees feeling lost on how to use AI in their workflows or how to deal with the fact that it can do most of their job for them, in less than half the time. While some speculate about wild futures where AI over-powers the world or humans conquer the solar system, it is clear that neither of those scenarios are optimal or totally feasible, functioning more as spectacles to incite critique (AI Futures Project, 2025).

As a designer that, like many others, feels intimidated and unsure about the future, I believe the solution is not to despair about the idea of human-replacing robots who will keep humans as pets, nor to believe that our brains will be fused with the cloud and that AI will rid the world of disease, war, and hunger. A more pragmatic approach must be adopted, where sufficient

regulations and guidelines are put in place to ensure AI works in our favour, to elevate human capabilities, and not destroy them (Makridakis, 2017). It is imperative that businesses learn how to exist in a world with artificial intelligence, generative AI, and LLMs, as in as the opposite could result in a weak workspace environment and cause a substantial drop in relevance in the market.

1.2 Aim

The aim of this research is to explore the challenges and opportunities of AI in the design field, how it is affecting designers and developers, and to propose a framework to optimise its adoption without undermining the work and value of human employees. The goal is to achieve a scenario where AI is seen as a co-author instead of a replacer or industrial manager, handling efficiency and data-driven automation while elevating the human workers' creativity, emotional intelligence, and strategic innovation, skills that AI cannot yet replicate (Bowles & O'Hanlon, 2025; Yu et al., 2025).

It is important to note that, in the context of this thesis, the title "co-author" describes AI as an instrument capable of generating, filtering and iterating design proposals and tasks within defined parameters. The term does not implicate consciousness of the machines but characterizes the technology as a partner rather than a simple servant, one that is capable of contributing to human ideas and following human established guidelines. The creation of meaning and authorship of the outputs are still attributed to the designer, but the AI's role in the process is being recognised (McGee, & Hedborg;.Vlaad, 2025).

1.3 Research questions

The main question: "How can AI be used as a "co-author" to improve design processes while retaining designers' agency?" is supported by two sub questions: "How are designers currently collaborating with AI tools in their everyday work?" and "How can processes be adapted or redesigned to support human-AI co-creation in organisational design contexts?".

1.4 The company

This research adopts an exploratory case study approach to ground the research goal "AI as a Design co-author" in a real-life organisational scenario and gain a deeper understanding of the current AI adoption landscape.

The company selected for this collaboration is Liip SA, a Swiss digital agency headquartered in Fribourg, which is focused on web development, implementation, and consultancy of digital solutions. With around 200 employees, referred to as Liipers, it stands in the middle point between a small and medium-sized agency. It is big enough to span six offices and have different project teams simultaneously, but small enough to feel its values across the organisation. Different from other agencies, Liip operates under Holacracy, a non-hierarchical governance model that follows a self-organised framework committed to agility, self-improvement, and the entrepreneurship of Liipers (Farkhondeh & Müller, 2021). By distributing decision-making responsibilities between employees, Liip ensures they maintain a certain level of autonomy over their work and are encouraged to take initiative. This structure is visible in the physical environment at the office, which is organised in an open space with different areas to take breaks, go outside, and spark casual conversations. It shows a clear sense of care over employees' wellbeing and the presence of the company's values that are supported throughout the office in the form of posters on agility, design principles, user experience, and the Holacracy model.

Liip's values are codified in their main principles that drive their decision-making and practice:

- **Purpose** over profits,
- **Open** over closed,
- **Challenge** over comfort,
- **Trust** over control,
- **Practice** over theory,
- **Flexibility** over strength,
- **We** over me.

Considering the governance model and together with these principles, Liip is a company that prioritises collaboration and experimentation, which is felt in their approach to technological change. Instead of imposing strategies or using AI as an internal driver for competition, the firm takes a collective stance in exploring different ideas, learning by doing, and involving different actors and disciplines in the process.

1.5 Delimitations

The scope of this research is limited to the design team at Liip, complemented by web developers with design-adjacent roles or experience. As a web and mobile agency, Liip's design

team is outnumbered by developers and works closely with disciplines connected to this collaboration, such as UX, web, and service design. Other domains present at the company, such as legal, analytics and business development teams, were excluded from the study as the research questions specifically concerns the impact of AI on the design practice.

The study is also bounded to an organisation that operates under a Holacracy model, a structure that is not widespread across the industry. This choice is deliberate as Liips' decentralised leadership model offers a relevant context to study human-AI collaboration in an environment ruled by individual autonomy. This research does not aim to provide generalised strategies for all organisational types, nor does it include methods to change or improve governance and leadership models as part of AI adoption.

Given AI's rapid evolution as an emerging technology, this research takes advantage of current knowledge and acknowledges that its accuracy may diminish over time as AI becomes more prevalent in the industry.

1.6 Contributions

This study contributes to Liip and other similar small to medium-sized enterprises with an intentional plan of AI implementation that they can adapt and integrate in their own strategies. Said plan addresses design and other creative teams, who demand different approach from tech related roles.

Beyond the practical framework, this research contributes to design research by reframing what the primary barrier to human-AI collaboration actually is. While existing literature identifies the challenge as the use of different tools and their integration in workflows, this study's empirical findings reveal that the more critical obstacles are structural and psychological. The erosion of designer identity, the perceived absence of institutional guidelines and the lack of psychological safety needed for responsible experimentation, are impacting the designers' adoption process and they question whether it is safe to try at all.

With this, AI adoption is addressed as a design problem in itself, one that requires co-design methods and human-centred strategies before introducing workflow interventions. The study further contributes a grounded understanding of how a Holacracy governance model both enables and complicates this transition, offering a relevant organisational context that is underrepresented in literature.

1.7 AI use

As a relevant exercise parallel to this research, different AI tools such as ChatGPT, Gemini, Nano Banana, Claude, and Grammarly were used during the study. These were used for tasks like finding niche references for specific topics, paraphrasing and correcting grammar, aiding citation making, explaining and connecting certain concepts and ideas, generating data maps, avoiding design fixation, and offering feedback and improvements to the text and results. All and any use of AI was carefully evaluated, fact-checked, reviewed, and edited as needed. The author takes full responsibility for all the content in this paper.

This experiment added a personal layer to my research as I struggled between accepting the efficiency the tools gave me and fighting for my and ideas over the machines feedback. Generated outputs constantly needed redirection to align with my thought process with was time consuming and highlighted the importance of prompting quality and continuous human input.

2. Theory

This chapter acts as the Research Clarification phase of the study. It presents research that maps significant moments in artificial intelligence development, including its benefits, challenges, and methods, as well as the history and relevance of codesign and service design, AI use in design, and AI implementation cases. The literature overview uses Google Scholar and Google search with papers ranging from 2011 to 2026. Most articles discussing AI are from 2024 onwards and older literature analyses previous predictions of the emerging technology and discuss other domains such as service and codesign. The research results from key searches including “AI adoption strategy”, “AI adoption case studies”, “AI use cases”, “AI in design”, “AI in codesign”, “AI implementation”, “AI creativity”, “Design using AI”, “AI anxiety”, and excluded papers that diverge too much from design and the goals of this study.

2.1 Designing with, not for: Co-creation in services

Design can be used as an umbrella term for multiple concepts. Most famously used to refer to the creation of appealing visual things like posters, brands and fashion pieces, design is more than making beautiful things; it is a problem-solving method that frames problems and provides unique solutions for a better future. Design thinking is an overall term used to encapsulate the

iterative process used by designers to make sense of and attempt to solve ill-defined and wicked problems with the best interest of the users in mind (Zamenopoulos & Alexiou, 2018).

Not all design disciplines have the goal of creating physical or digital products; design can also be used to develop and improve human-centred policies, processes and experiences. Service design (SD) is a collaborative approach, focused on organising different stakeholders, providers and actors to achieve the best version of a service that meets the users' needs and generates value for all people involved (Steen, et al., 2011). SD is a multidisciplinary subject that combines expertise and methodologies from diverse areas such as marketing, cognitive science, engineering and user experience. This broad skillset allows designers to identify meaningful problems, frame them effectively to explore different opportunities and find solutions that would not be possible without this holistic mindset (Figueiredo et al., 2024; Stickdorn et al., 2018).

A customer's experience of a service or a product can be almost more important than the product itself. A service that solves the user's needs but leaves them frustrated or underwhelmed can ultimately leave a negative impact on the customer's opinion of said service. For example, an unfortunate interaction with a nurse at a hospital may lead a patient to choose a different medical facility in the future, even if the treatment itself was successful and the illness was cured (Stickdorn et al., 2018). The importance of service design is increasingly prominent in a world where everyone shares their opinions, experiences and grievances publicly, and companies can be put in the online spotlight at any given time. Corporations looking to innovate are investing more on improving their services, not only externally but internally, catering to their own employees' needs as well as their customers (Stickdorn et al., 2018).

Creating a service that caters to everyone is not an easy task and may even be virtually impossible, but that does not mean it should not be strived for. Welcoming the users into the design process as collaborators brings a whole different perspective to what is normally a team of designers or engineers conducting some user-testing but mostly doing the work by themselves. Co-design, meaning collaborative design, is an approach that combines a variety of methods where different actors come together to co-create something (product, service, idea) that will benefit as many people as possible. It is a powerful tool to explore wicked problems that cannot be resolved by a single team or person as they are too complex and do not present straightforward solutions (Coyne, 2005). Co-designing involves exploring the current situation people are living through, defining pain points and opportunities and imagining a shared desirable future.

This is achieved by identifying who and what can help them create that future, and what the next steps are that need to be taken (Zamenopoulos & Alexiou, 2018).

In a service design context, codesign is an extremely important aspect of the process since it works as a common language for a diverse group of people. Through the creation of boundary objects like journey maps, mind maps or service blueprints, different participants, often from a variety of backgrounds, can reach a common understanding of the issues and experiences of one another (Stickdorn et al., 2018).

From an organisational standpoint, using a service design approach and codesign methods to improve a service brings a multitude of benefits, including increasing customer loyalty, reducing costs, optimising decision-making, improving collective creativity and innovation practices, boosting efficiency and effectiveness, thinking outside the box and allowing continuous upgrading (Steen, et al., 2011).

When talking about codesign in artificial intelligence, it is relevant to go back to one of the roots of co-design: Socio-technical design. Originating in the 60's in post-war Scandinavia, it came to be thanks to the industrial democracy movement with the belief that workers should have direct involvement in the management of their own industry, not only empowering the workforce but improving productivity and efficiency. The emergence of personal computers in the 1970s and their implementation in different industries pushed two Norwegian workers' unions (from the Iron and Metal industries), to come together in a participatory design project with the goal of designing how and when computers were going to be used in their jobs. It gave employees the power and freedom to influence their own workflows alongside their managers and consulting engineers, serving as a clear example of a group sharing their own experiences, knowledge and resources to create a better situation for everyone involved (Zamenopoulos & Alexiou, 2018).

In today's environment, plagued by constantly emerging technologies and rapid technological advancements, it is becoming harder and harder to find solutions for every problem. Between the fear of falling behind and the consequences of using AI with no clear plan, ethical AI adoption is a goal with too many variables and connections to be considered a defined problem. So, just like Norway's participatory design project on computer use, the deployment of artificial intelligence in the workplace deserves to be considered by all members involved, the employees, the engineers, the bosses and managers, legal teams and designers (Bowles & O'Hanlon, 2025). The democratisation of this adoption process requires allowing the participants to come

up with their own solutions, work side by side, brainstorm with their colleagues and open space to develop new skills (Zamenopoulos & Alexiou, 2018). With that, codesign will be an indispensable tool to understand, analyse and explore what AI-powered business and services look like, and reaching a truly human-centred AI environment will imply extensive collaborative projects to be conducted.

2.2 Demystifying AI

Humans have been trying to predict the future since the dawn of time. We have imagined flying cars in floating cities and living on Mars, but also the dark realities of intelligent robots turning against humans, craving for power and using their abilities to replace every remnant of humanity on the planet. Science fiction is a staple of human creativity, but it also emphasises our deepest fears and doubts about the unknown. With recent advancements in technology, these realities may not be that far away from us.

On one hand, there is the utopian scenario that would unfold with humans being truly free, not chained by jobs or responsibilities that would instead fall onto AI-driven robots. Technology would be able to erase all disease, people would be connected through the cloud, where all our brains would be uploaded to, and nanotechnology would create anything and everything we would need. On the other hand, there is a real possibility that unchecked innovation will result in machines ruling the world, making major decisions and turning people away from their jobs (Makridakis, 2017).

Nowadays, artificial intelligence seems to have taken over the world, transforming everything from back-end programming to marketing and social media. Every website and application now has a new AI agent or functionalities. From translating videos on YouTube and automatically editing pictures on Google Photos, it even summarises emails and offers message suggestions to our conversations, AI is everywhere. On the downside, this has led to a higher percentage of the population being concerned about job replacement than excited about AI's capabilities. Different reports show that up to 47% of jobs could be automated by 2033 (Ramaswamy, 2017 as cited in Prentice et al. 2020), especially in industries related to data collecting and processing, which would cause a noticeable amount of job loss because of the technology (Prentice et al. 2020).

It is important to note that the AI mentioned in this research is referencing Artificial Narrow AI, as in use cases of AI across industries, and not the hypothetical artificial general intelligence

capable of outperforming humans in any task. A main component of artificial intelligence is machine learning (ML), which gives computers the ability to analyse and comprehend data through pattern recognition. This usually implies deep learning (DL), a subset of ML that uses several layers of simulated neurons connected to loosely mimic the structure and functions of the human brain in a virtual setting. Before deep learning, neural networks typically had only three to five layers with dozens of neurons, in contrast with deep learning networks that contain millions of neurons across dozens of different interconnected layers (Chui et al., 2018; Cui & Hung, 2025).

It is hard to find a corner of the world completely untouched by this new wave of technological marvel, but contrary to what many believe, artificial intelligence is not a recent invention; it has been around for many decades.

The concept of creating artificial neurons goes as far back as 1943 when Warren McCulloch and Walter Pitts imagined a computational model of our neural network. Throughout the decades, multiple others expanded their ideas, and the networks became increasingly complex and powerful, especially regarding image recognition. All these years of research eventually led to reinforcement learning and generative adversarial networks that are used today to continue training more advanced models to do tasks like understanding images, language and text, decision-making, and more importantly, learn from experience without the need for human intervention (Chui et al., 2018).

While machine learning offers a large range of problem types, some are more pertinent to design practice than others. Classification and clustering analyse and organise qualitative and quantitative data, offering a clear synthesis for designers. Anomaly detection can identify irregular user behaviours; ranking and recommendation allow for the personalisation of digital products and experiences, and data generation is reshaping the design process through innovative generative models (Chui et al., 2018). Large language models (LLMs) such as ChatGPT and Gemini operate across many of these problem types, providing everyone with easy access to human-like conversations, image generation and data analysis.

Artificial intelligence brings more advantages to specific tasks that involve algorithms, data and numbers. For that, it performs best in industries with copious amounts of data, including robotics and programming, and minimal restrictions on its collection and use. In reality, LLMs and other AI models do not create new information but source datacentres for content and output an amalgamation of a lot of different inputs, posing a risk to copyright, privacy and

ethics. On the other hand, creativity, emotions and contextual understanding have not yet been successfully translated into algorithmic systems and have a long way until they reach human-likeness (Dwivedi et al., 2021). As AI technologies become more evolved and accessible every day, it is crucial that designers understand the fundamentals of these tools to be able to critically evaluate their true capabilities and limitations in a creative context.

2.3 Using AI as a co-creative

Everything is designed, from clothes to posters and social media marketing, websites and apps, video games we play and kitchen appliances that we use every day. There are creatives behind almost everything in our lives, catering to our wants and needs (Chanchamnan et al., 2022). Although they belong to different disciplines, all design practices share something in common: its process. The design process involves engaging in critical design thinking in order to break down complex problems, creating new ideas and applying them to practical and grounded solutions (Yu et al., 2025). It is a deeply human technique that implies contextual awareness, empathy and curiosity for others, an acute emotional intelligence and of course, creativity.

Now, with the convenience of AI-generated images, poems and songs, some believe that human-made art is in peril. Creativity, a trait always regarded as uniquely human, is now being artificially mimicked to create "new" content based on limitless amounts of data from real artistic work made by people. This has prompted countless debates about the true meaning of creativity and originality, especially when considering ownership laws and copyright. Aligning with Garcia (2025), this study defines creativity as the ability to produce original, authentic and intentional ideas. In other words, truly creative ideas need to be more than practical, they should be meaningful and contextually aware, qualities that are not possible to achieve using AI's patterns-based generative models. Furthermore, this study also aligns with Boden's (2004) typology of creativity, recognizing the combinatorial capacity of AI models to merge different referents to create something new, while also understanding that it cannot indulge in exploratory, and transformational creativity. Due to the lack of intentionality and meaning creation, an artificial agent cannot generate nuanced results that go beyond superficial visual aesthetics and is only capable of assisting humans in basic creative tasks while they conduct the exploration.

If the belief that AI art will indeed replace humans and that people will become merely "arbiters of creativity" (Garcia, 2025, p.2) turns out to be true, eventually "art" will stagnate, as genera-

tive models currently need new inputs to be able to create new works. Without new data, generative models would have to scrape other AI-generated art and consequently produce weird results, devoid of any emotion, purpose or conscious decision-making, losing the human aspect of art (Garcia, 2025). This is not to say that AI cannot be used in the design process, but it requires "a delicate balance between embracing technological advances and preserving the essence and appreciation of human creativity" (Garcia, 2025, p.5).

Of course, not everyone has this essential sensibility and "AI slop" keeps getting churned out on social media and in the physical world. Slop, low quality content that is mass produced with little to no oversight, presents itself in pictures of Jesus made by shrimp on Facebook, fruit babies eating other fruits on Instagram, nonsensical restaurant menus, creepy advertisements with people with too many hands, musical albums on Spotify with the same song over and over, and inaccurate published articles on apparent legit journals (Resnik & Hosseini, 2026). The infestation of our daily lives with AI slop, regularly posted without being transparent about the clear AI use, is tarnishing the public image of AI technologies and hindering the possibility of creative workers even considering using it in a smart way.

Many non-artists using image generators like this to cut corners and not hire designers, or going as far as creating and selling AI art that is based on real artists' works, pose a real threat to their creators' livelihoods and well-being. This has put a big taboo label on using AI as an artist, as it usually results in backlash from their peers, feelings of guilt and responsibility for the crimes committed against the art community, perpetuating bias, contributing to pollution and taking part in other moral dilemmas (Garcia, 2025).

It is also interesting to recognise that the majority of deep learning tools currently available at our disposal are drawn and created by engineers and computer scientists who use a contrasting approach to designers. While engineering design tends to focus on converging ideas according to the initial problem and all its requirements and rules, designers begin their process by understanding the problem space and drawing a variety of divergent concepts. This gap definitely influences how people perceive AI and its uses, usually relating it to rational and mathematical tasks and struggling to imagine it in the messy and heuristic nature of creativity and imagination (Lee et al., 2025).

To truly use AI as a co-author in art and design, it would ideally showcase generated answers similar to humans' and present a consistent understanding of emotions and their regulation. In Schlegel et al (2025)'s study, LLMs have shown positive results in emotional intelligence (EI)

tests, even showing higher scores in the Levels of Emotional Awareness Scale (LEAS) than the average person. These tests, although simple and not a true representation of real worked interactions, included recognising others' emotions, describing feelings, assessing and creating scenarios according to the emotional state of other people or one's own while considering the consequences of these courses of action (Schlegel et al., 2025).

Taken at face value, it may seem like artificial emotional intelligence is the key for AI-generated art and that LLMs can in fact generate something that is emotionally impactful, but in reality, this factor of the technology is in its early stages, and the results can vary tremendously between prompts and different contexts. As mentioned in the literature, while LLMs excel in these performance tests, emotions, expressions and body movements are all dependent on a variety of aspects of the human experience, including culture and ethnicity, fatigue level and mood, which AI is not yet trained to discern. It is also interesting to mention that the scenarios created by artificial intelligence lack diversity compared to its human counterparts, and that the limited transparency of these models makes it harder for future improvements to be implemented since no one has a concrete idea of how they work internally and how these changes could affect them (Schlegel et al., 2025).

Emotional intelligence is one of the most valuable traits to have when contemplating AI adoption. More than identifying and understanding emotions, EI gives people the ability to manage good and bad situations and reflect on their own feelings and experiences. People who are highly emotionally intelligent are prone to have healthier relationships with family, friends, and romantic partners, report having more career success, and in general, tend to live a happier life. From an industry perspective, emotionally intelligent leaders are more motivating and good decision-makers, and employees are more committed to their roles, resulting in better performance and job satisfaction, which impacts their businesses for the better. It has also been found that highly educated employees see AI in a positive light and expect it to improve tech-related aspects of their work (Morikawa, 2017, as cited in Prentice et al., 2020; Singh et al., 2024).

In order to successfully implement AI in service settings and companies, it is necessary that both the employees and the employers have a high level of EI, so as to have a critical approach to where to implement it, how it will influence work, and impact users. Customers still prefer to interact with other people and not be met with an uncanny AI voice on the phone (Konuk et

al., 2023), and that requires empathetic designers to make sure this need is met (Kaur & Sharma, 2021; Prentice et al., 2020).

It is clear how EI takes part in the design process. EI improves group work (Prentice et al., 2020; Singh et al., 2024) and makes people more open to others' experiences, which are the basis of service and UX design projects. Creating user-friendly solutions involves high levels of creativity, user research, and concerns over usability and accessibility, things that come naturally to designers but AI struggles to reach on its own.

Some may claim that "AI is the new UX" while in reality it should be used as a powerful tool to help designers create highly customizable and adaptable user-centred experiences. (Chanchamnan et al., 2022). Using AI as a co-author then implies always having the user's experience in mind, especially when referring to UX and service design but also other areas like graphic design and fashion, where it can use data analysis to guide the creation process towards a more universal and inclusive outcome. The goal should always be to improve processes, services, and products and not overcomplicate the creatives' lives. For this, it is good to take advantage of already developed and tested AI tools and not lose so much time exploring a lot of new up-and-coming ones that can become overwhelming, even though both are useful (Chanchamnan et al., 2022).

So, what exactly is the design process? Design is not just an end result; it starts even before initial sketches with the definition of the problem. Here the focus is on research and defining the goals and constraints of the project. It is then followed by the ideation stage where the objective is to generate a variety of different ideas and explore many possible solutions to the problem. The feasibility of these ideas is assessed in the evaluation stage that happens simultaneously to the ideation, in what can be called the creation stage. In this iterative process, or "mode-shifting" (Lazar, 2018 in Yu et al., 2025, p.2), the creators switch between creating and judging ideas to work towards a concise and secure solution. It is in this phase where methods like user testing and prototyping happen before the idea is ready to go into production in the final outcome stage.

Another way to see the design process is through the Double Diamond model, a framework that is often used as a base user-centred design. While it is a simplification and real design work would include many more layers, nuance and steps, it serves as an overall example of the essential phases that happen in a project. Like the previous description, it is also divided into four parts: Discover, Define, Develop, and Deliver (see figure 1).

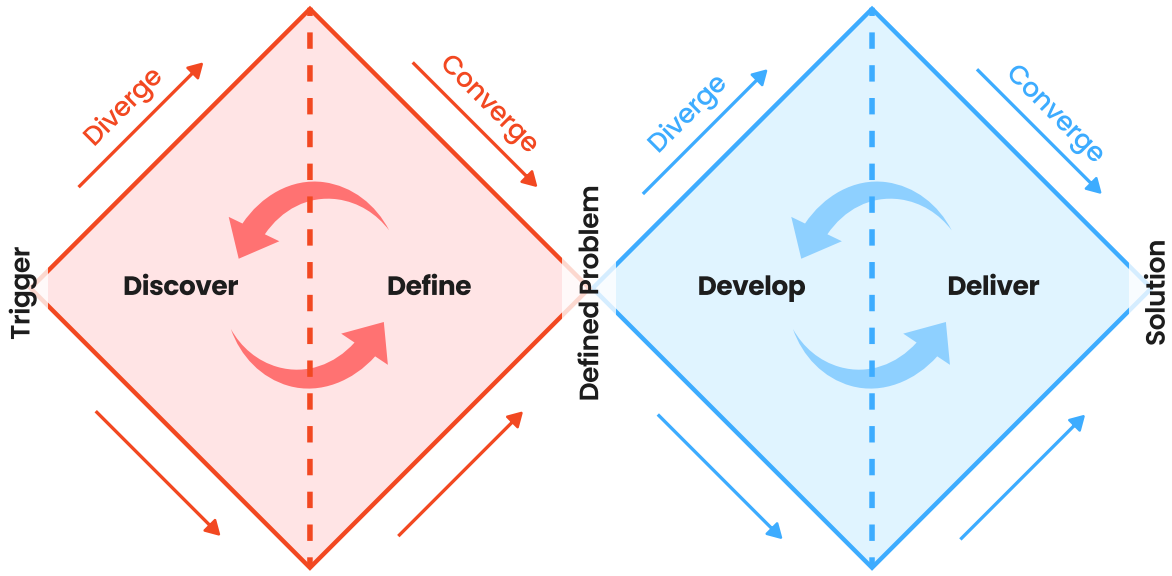


Figure 1 - Double diamond design model adapted from the UK Design Council

With this model, we can clearly see what each phase looks like: the research makes the divergent section of the problem diamond as designers gather a lot of different information and case studies about the context and the users' needs, which they then converge into a concise framing of the problem and its characteristics. In the solution diamond, designers diverge once again with the development section, or in other words, the ideation phase, which converges as they test and prototype different ideas. In this model, the creation phase is contained in the second diamond, and the outcome stage is simply the result of all the sections and can be found on the point furthest to the right (Lee et al., 2025).

When applying artificial intelligence tools to the design practice, it is necessary to consider all these different phases and not assume that the only relevant outcome of the creative process is the end product. Between using data analysis and pattern recognition to aid research and creating interactive personas for user testing, there are a lot of underexplored possibilities of neural networks in design that can complement designers' skills and lead to innovative change (Lee et al., 2025).

2.4 Challenges and opportunities of AI

Now that the deep learning methods have been contextualised in the design environment and designers' valid apprehensions towards artificial intelligence have been acknowledged, we can take a deeper look into the reality of these tools. What are the actual issues, how can they be used in reality, and what are the responsibilities that come with their use?

2.4.1 Challenges

The introduction of AI assistants and AI-powered programmes in the workplace comes with increasing expectations. Instead of giving workers fewer working hours and more free time for leisure, it expects them to do double the work at twice the speed. The obsessive focus on productivity and efficiency comes at the expense of the employees who are the target of a command-and-control management mode. By using deep learning to monitor workers' behaviour and manage their performance, from mouse clicks and key presses to verbal responses, email tone, and even break times, companies are taking advantage of digital tools to impose an environment of "digital Taylorism". Evocative of the Industrial Revolution's era of Taylorism, the workforce is managed to an extreme to standardise the labour process and optimise the company's operation, even if it costs a good working environment and employee autonomy, freedom, and wellbeing. The reality is that even if it results in a short-term gain in overall productivity, overusing algorithms in hopes it will increase work rate will only destroy the human value, a loss that is bigger and more impactful to the industry. If workers' mental health steadily declines, so will their proactivity and loyalty to their workplace, devaluing the company (Bowles & O'Hanlon, 2025; Konuk et al., 2023).

AI is not a be-all and end-all solution for businesses, and its use without care and investment in human capital makes it simply an expensive gimmick and can even become unsettling. Although automation can bring significant benefits and simplify a lot of tasks, it tends to undervalue human traits like curiosity and empathy that are key for driving innovation. Tacit knowledge like intuition, talent, context awareness, and emotional intelligence cannot be replaced artificially and are some of the fundamentals for successful leadership and work ethic. Using innovative technologies does not intrinsically mean someone is being innovative if it is used wrongly and contributes to the present unresolved issues underneath. If the goal is to promote productivity, forcing automation is not the way to go, as it should instead focus on collaborating with employees, fostering their skills, boosting their knowledge about emerging technologies, and developing their long-term success that will consequently bleed through to the firm's overall performance (Bowles & O'Hanlon, 2025). Even for many users, the overwhelming personalisation of chatbots, websites, and other experiences has become unsettling (Bughin, 2025).

The overarching issue regarding AI adoption can be explained in simple terms: a lack of strategy. There are an abundance of issues that will not be resolved so soon and cannot be fixed by

everyday people. While the responsibility falls on the AI companies and engineers to find solutions, we have the power to design our own strategies to mitigate them the best we can. Most available large language models and media-generating AIs are still premature technologies that go through constant updates, and it is not realistic to expect them to reach perfection and not cause any issues in the future (Wu et al., 2021). The immaturity of the technology only serves to highlight the importance of organisations drawing their own strategies from personal experience, considering there are no established methodologies. Better said than done, it is no surprise that AI implementation remains with a high failure rate, especially in small and medium enterprises (SMEs), thanks to the lack of proven guidelines and all the risks that emerge from it (Al-Kfairy, 2025; Madanchian & Taherdoost, 2025; Sánchez et al., 2025).

Lacking a plan can significantly affect both the business and human side of an organisation. Humans will naturally resist technologies that challenge their routines and ways of working (Dwivedi et al., 2021), especially if these are brand new and are forced onto the workplace abruptly. Not only do people fear the unknown, but they also still cannot perceive the full benefits of these automation changes since they usually imply long-term outcomes or intangible gains like “more inter-team collaboration” or “happier employees”. This lack of visibility often causes anxiety and distrust for both the tools and the leadership who are supposed to guide and protect their human workers. For the company, this translates into inertia and a lack of innovative mindsets that can keep companies back from improving their businesses and falling behind their competitors (Merhi, 2023; Sánchez et al., 2025).

A vague strategic plan can lead to malfunctions, dissatisfaction, lack or misuse of resources, etc. (Merhi, 2023). The goal cannot be just “automate everything”, a common misconception that is propagating unrealistic expectations about AI technology. It requires a careful selection of tasks that are capable and ready to be automated, consider the level of automation and human involvement necessary for said tasks, manage the impact of automation on the human workforce, and draw contingency plans in case of errors the AI will inevitably create. These decisions do not have a rule and must be made according to each company's environment, people, and goals (Dwivedi et al., 2021). In the design context, it is even more important to establish guidelines and best practices as creative tasks cannot be fully automated, and the results of the design projects will most likely directly impact its users (Garcia, 2025). At the same time, contrary to areas like programming and finance whose work is usually hidden from the public and in turn more acceptable to be AI-powered, designers struggle to convince their higher-ups to invest in and adopt design-centred AI tools, considering the large amounts of data and time

needed to train models for design tasks and the high price tag that would come attached to it (Chanchamnan et al., 2022).

Responsible AI systems are very expensive (Bughin, 2025). Considering that many organisations in the industry already struggle with regular system updates (Merhi, 2023), we can understand how the development of AI infrastructure requires financial capabilities that not everyone has access to and that consequently limits who can begin to implement these technologies (Dwivedi et al., 2021). Usually, this leads smaller enterprises to opt for ready-to-use AI solutions like simple chatbots rather than investing limited resources in a customised framework (Sánchez et al., 2025). Moreover, constant updates must also be accounted for in the budget after the initial enhancement since the technology is rapidly evolving and is hard to catch up to, including acquiring new and skilled personnel. Bigger corporations with a larger resource pool will have a clear advantage over late entrants who will only be able to participate when the prices eventually drop to invest in the automated industry (Al-Kfairy, 2025).

From an organisational point of view, there are bigger obstacles than others. The implementation of an AI system is a complex process that requires a lot of infrastructure and resources from the companies. A stable hardware and software, as well as adequate storage and servers, are necessary for the success of integrating new technology and facilitating communication. With inadequate established IT structures, not only is integrating new generative technologies extremely intricate, but the specialised workers who can successfully do it are hard to come by. Older systems do not always interact well with newer additions and need significant modifications that can possibly disrupt existing workflows. Such changes are considerably costly both financially and timewise, which businesses should deeply consider before deciding to follow the AI adoption trend, since it is one of the major factors of failure in this endeavour (Al-Kfairy, 2025; Merhi, 2023; Sánchez, Calderón, Herrera, 2025).

Data collection is one of the biggest challenges of adopting AI, simply for the outlandish amount that is needed for training the models, and the requirement it must fulfil to ensure accurate decision-making (Merhi, 2023). It must be comprehensible, reliable, valuable, varied, and quick to collect. It also should be as transparent and clean as possible to avoid mistaken assumptions and train the network on wrongful data, as the quality of data determines the success of the AI (Chui et al., 2018; Dwivedi et al., 2021; Sánchez, Calderón, Herrera, 2025). Even with a set strategy for data collection, integration, and processing, firms also need to account for “overfitting” and “underfitting”, where deep learning models either learn from the datasets

too well and will struggle performing based on new data, or fail to recognise any patterns and compromise its performance independently of the data it is trained on. For that, a lot of manual labour must go into guiding and supervising the learning process by labelling training data and analysing inputs (Chui et al., 2018).

To further complicate this process, deep learning data visualisation tools often need to be used to make sense of the data and help people reach a mutual understanding of the situation and improve interpretation of the numbers and insights collected. Without this assistance, the work done by the neural networks is quite often incomprehensible, adding to the complexity of AI systems (Dwivedi et al., 2021).

The mass collection of data comes with its own security issues (Wu et al., 2021) that entail having strong security protocols, data usage policies, and data sharing regulations between roles and/or institutions (Chui et al., 2018; Dwivedi et al., 2021; Sánchez, Calderón, Herrera, 2025) to prevent misuse, theft, and fraud (Merhi, 2023). These include the encryption of the da-ta, employing secure storage guidelines, and making sure no sensitive information is collected without a specific reason in the first place. If not followed, leaks and security breaches can severely impact businesses' reputations and lead to long and costly legal repercussions (Al-Kfairy, 2025). It is imperative that stakeholders are confident that their data is protected and will not be lost or sold, but putting these protocols in place takes large amounts of time, planning, work, and a lot of bureaucracy that might influence some firms to skip this step.

Not only is the security of the data essential, the initial pool of data must also protect people's work and respect copyright laws. That being said, this regulation is not currently enforced and can lead to many issues, like posing a danger to artists and creators who have showcased their projects online or even resulting in national security breaches and other threats from foreign organisations that get hold of confidential information (Dwivedi et al., 2021). Ownership has become a debated hot topic, with some defending the transformational power of AI and the merging of various influences and others fighting to preserve intellectual property and everyone's right to privacy and data security. The truth is that there exists a gap between these issues and the current copyright laws based on human authorship, and while legal institutions try to resolve this intricate situation, our reality continues to lie in the grey area where AI companies and regular users can get away with using original works without permission and the original creators cannot get their warranted credits. Although artificially generated works are also not

protected under copyright laws, they still cause considerable uneasiness to designers, painters, videographers, photographers, and other members of the creative industry (Garcia, 2025).

The requirements for the data collected go beyond just quantity; the quality is extremely crucial. Like mentioned earlier, good quality and relevant data directly impact the quality of the model and prevent risks like hallucinations, bias, and prejudices. Artificial networks do not just adopt our rationality and language; they source all human culture, including the moral ethics and judgements of its human creators, without fully understanding them. AI models struggle to understand meaning, context, and nuance (Bowles & O'Hanlon, 2025). When the data comes from heteronormative and monocultural contexts, it continues to reinforce outdated biases and social profiling that reinforces gender roles, spews racist and sexist rhetoric, and creates homophobic and transphobic concerns, all targeting groups that are underrepresented in the training data. While it does not have access to more diverse and accurate details, it will not be able to move past these "beliefs" or suddenly gain the capability to reflect on them (Chui et al., 2018; Dwivedi et al., 2021).

In an anecdote from Amazon, an automated recruitment tool actively turned away women's résumés because it was trained based on male curricula (Bowles & O'Hanlon, 2025). This has not been the only case of sexist algorithms; a recent collective experiment showed that LinkedIn has a preference for male profiles over females, and while the company denies it, it is felt across industries, and women reported a significant difference when they changed their profile name and picture (Down, 2025). In other cases, X's (formally known as Twitter) initial experimentation with conversational agents resulted in hateful discriminatory content being posted before the first day reached its end (Dwivedi et al, 2021), which resembles the current situation with Grok, the website's recent chatbot, where it labelled itself "MechaHitler" and delved into a serious antisemitic rant. This all culminated from a directive by the team behind the model, instructing it to avoid spreading politically correct posts "as long as they are well substantiated" (Hagen, 2025). Well, it was substantiated on hate speech frequently supported by Elon Musk himself (X's owner), a clear act of the weaponisation of misinformation.

In other cases, unrelated to discrimination, algorithms trained on past data can create inaccurate forecasts based on irrelevant or made-up concerns, especially in constantly evolving markets like sales. This salients the importance of not becoming complacent or, in other words, falling for automation bias, where humans believe that the AIs are generally better than them and do

not need to be fact-checked, prompting commission errors— where a human follows an incorrect recommendation from an algorithm— to happen (Dwivedi et al, 2021).

Since most LLMs are researched and developed in Western-centric settings, the same expressions, rules, and meanings it is trained on may not translate well to non-Western populations (Schlegel et al., 2025). Likewise, something similar can happen in the art context. When a generative model is predominantly trained on Western art, it will reflect the same views and themes of the Western world, marginalising all other traditions and creative expressions that transmit different points of view (Garcia, 2025). Some may take the resulting outputs as proof that some artistic styles are superior to others and use it to defend misguided ideals and xenophobia.

How artificial intelligence works internally is still a mystery to us. Besides the initial spec given to the model by its creators, the algorithm's process of decision-making and generation of new data remain hidden (Merhi, 2023). This "black box" phenomenon causes a lack of transparency and honesty that severely impacts users' trust and prevents the faster adoption of this technology (Bughin, 2025; Dwivedi et al., 2021; Garcia, 2025; Schlegel et al., 2025).

Plagued with inaccurate data and a lack of transparency, large language models begin to hallucinate. Hallucinations are characterised by fabrications by the AI, usually as a response to a question it did not know the answer to but also observed in otherwise regular, everyday generations. According to the literature, up to 25% of generated summaries contained hallucinated content (Ahmad et al., 2023). Both image generators and large language models can fall ill with this phenomenon, creating media that spreads disinformation and manipulates and distorts facts (Garcia, 2025). Furthermore, these models find it challenging to admit that they are wrong and often resort to creating more fake information to defend their previous made-up answers. Dangerous content like this brings up concerns about accountability over content generated by artificial intelligence. Is it the responsibility of the user, the AI developers, or legal institutions to prevent and mitigate these types of situations? (Chui et al., 2018; Garcia, 2025; Merhi, 2023).

There are multiple factors that contribute to hallucinations. Like previously mentioned, the use of unreliable and biased data is the most glaring one, but even with completely accurate data pools, the process of combining information from different sources can result in the creation of false statements anyway. In addition to that, the lack of nuance and context, plus the lack of sequential reasoning, leads to incoherent responses and self-contradictions (Ahmad et al., 2023).

Hallucinatory behaviour is incredibly dangerous in certain high-stakes environments like healthcare, politics, security, and even some user research projects, in which the absence of critical thinking and double-checking all the information can bring huge risks to patients, users, and the general population (Ahmad et al., 2023). The manipulative ability of LLMs has been taken advantage of before in cases of election interference through unauthorised data use (Bughin, 2025), fake images about terrorist attacks, and fabricated fear-mongering news with the goal of causing chaos among the public or the benefit of certain individuals (Garcia, 2025).

With all of these risks of using generative tools, it is understandable why so many people choose to avoid them. Of course, they can be used correctly in a way to mitigate these errors and abide by ethics and morals, but that implies training how to effectively prompt these models, comprehend what the limits are between what they can be tasked with and not, and having the reflex to always double- and triple-check the results. None of these are common knowledge and take time and effort to perfect. The social pressure on employees to quickly adopt new tools and the competitiveness that it cultivates cause workers to suffer from "AI anxiety" and "Learning anxiety" that can significantly affect their mental well-being (Bughin, 2025; Konuk et al., 2023). Without skilled human experts that have the skills to implement, manage, and work with AI technologies both in a technical and creative aspect, businesses are forced to undergo major internal adjustments to roles and team structures, which impairs a smooth AI adoption (Merhi, 2023; Sánchez, Calderón, Herrera, 2025).

Resistance is usually not about the technology itself but the consequences that the new system will bring (Merhi, 2023). In practice, artificial intelligence is capable of automating a lot of different tasks, especially in data-heavy industries that require maths, data collection and classification, programming, etc. Moreover, it is predicted that up to one third of current positions could be affected by automation by 2030. The biggest impact of automation will probably be felt in emerging economies where a lot of the population finds themselves working in industries with repetitive tasks that do not involve a lot of skills. Taking into account how marginalised groups, including immigrants and women, are often the ones working in these areas, job displacement can cause critical damage to their well-being and means to live. In contrast, new skilled jobs are expected to appear in developed economies that will partly compensate for their job displacement, extending even more the class divide and inequality in the world (Dwivedi et al., 2021). Those who have the opportunities will search for new skills and knowledge, while those who cannot will perpetually be left behind. The expectations for new workers keep getting higher since a smaller number of people are now expected to work with machines and do

triple the things they did before. Fewer people with more tasks seem to be the goal for a lot of companies looking to maximise their productivity, taking no consideration for the livelihood of its workers or the quality of the job.

Job insecurity is one of the biggest anxieties related to the AI adoption phenomenon, and many people are unsure of what their future careers will look like. With how quickly the technology evolves, university degrees are outdated by the time students finish their courses, and everyone is struggling to keep up with the demands of the industry. Even if replacement does not end up happening, the use of automation makes people dependent on technology to do their jobs, which translates to a loss of agency and autonomy. People slowly stop reflecting on what the AI is giving them and start relying too much on it to do every single task, even cognitive ones (Bowles & O'Hanlon, 2025). Often without noticing, the overreliance on the technology leads to a decrease of people's situational awareness and manual skills needed in case of failure which becomes increasingly more probable ad the AI becomes more intelligent (Endsley, 2023). In creative fields, over-reliance on AI outputs will negatively impact both the design process and its results. If instead of using artificial generation tools to augment their own creations, designers use AI to make a concept and image for them, they effectively lose creative control over their project. By depending on technology to be creative for them, artists will limit their own expression, motivation, innovative thinking, problem-solving skills, and creativity (Yu et al., 2025).

Since technology is not perfect and is bound to fail occasionally, human tacit knowledge is still essential in case of failure or need for adjustment. If the human workforce believes the algorithms are always right and stops being critical towards the outputs, there will be serious consequences when the technology is down, and no one knows what to do and how to do it. This serves to once more highlight that the solution is not to replace people but elevate the human value and augment their skills, rather than making them lazy (Dwivedi et al., 2021).

In a design environment, these issues are felt even stronger. Creatives are pushed to adopt new technologies to secure their jobs and avoid facing obsolescence, with a lot of enterprises seeing AI as a way to cut corners in the creative department (Garcia, 2025). Since image generators can create design assets for free and in a couple of minutes, they can avoid the need for weeks of research and development from the design teams. AI slop continues to be a threat to designers when authenticity loses its value on the market against quantity, and it essentially dilutes

traditional artistic skills and intentionality. The mass production of artificially generated designs oversaturates the market with low-quality products and harms both clients and businesses. This, of course, happens when leadership does not value artistic labour and quality, lacks foresight in business planning, or simply cares more about monetary gain and hopes that regular customers will not notice the difference between human and AI-generated designs.

Because artificial intelligence development has been led almost exclusively by engineers and computer scientists, it is expected that the biggest AI platforms benefit engineering practices and data-related tasks. Design and the arts have been largely overlooked in this development phase, thanks to the inherently analytical nature of neural networks and the absence of creative stakeholder teams in the creation of these systems. AI models famously struggle with complex emotional tasks compared to simple and direct prompts, and many tools are not equipped with design-specific language and terminology that impedes designers from being able to accurately transmit their goals to the machines.

Schön (1992) identifies several aspects of designer know-how that machines would need to replicate to truly comprehend how design is practiced. The ability to see, both literally registering visual information and simultaneously judging its quality, allows designers to determine what needs to change or be developed further. This skill is not only directly connected with intentionality and individual perception, but it enables designers to recognize unintended consequences. That is, results that were not initially intended but reveal new qualities when engaged with. Pattern recognition is another important skill. Designers perceive figures, groupings, and spatial relationships in marks on a page through gestalt processes that Schön argues computers fundamentally struggle reproducing. Additionally, design is also described as a sensory practice, where engaging with materials leads to the construction of design worlds shaped by subjective appreciations of the process and the behaviour of the materials themselves. Finally, prototypes are highlighted as instruments of reflection and change that transform according to the context and help creatives define the rules of the design. Without the capacity to enhance designers' "seeing" and know-how expertise, or expanding their "repertoire of prototypes", artificial intelligence falls short of becoming an adequate assistant of design practitioners.

This gap leaves designers frustrated with the absence of AI support for ideation, exploration, and prototyping in the design process and binds them to the existing inadequate tools. It is hard for them to imagine the possibilities that new, innovative tools could bring, as they often do

not understand all the capabilities of AI technologies, and trying to prototype a new AI tool is a challenge in itself. The unpredictable nature of algorithms and the fact that they need to be retrained for every single use case and change in circumstances stand in the way of innovation in its applications in creative areas (Chanchamnan et al., 2022; Chui et al., 2018; Schlegel et al., 2025; Yu et al., 2025; Zhang et al., 2023).

Being a mostly visual medium that calls for creativity, artistic sensibility, communication, and emotional skills, there is no right method to insert AI into the creative process. It is up to each designer to go through a trial-and-error approach to learn new skills and find unique ways to augment their creative vision. This tedious process presents a steep learning curve that functions as a barrier for many creatives (Yu et al., 2025). There are so many niche tools available besides giants like ChatGPT and Nano Banana, and understanding the parameters of each individual programme poses a deal-breaker when trying to use AI as a design co-author (Zhang et al., 2023). Every platform has different functions and languages that make it hard for designers to quickly change between them or try a new one, since it would take a lot of time. It is not enough to ask AI to make something new; creatives need to be able to transmit their unique ideas to an algorithm that struggles with creative tasks. After that, they need to fix most of the outputs and go back and forth until it reaches something they are happy with. A lot of the time, the situation ends up with either designers accepting subpar results or coming to the conclusion that, at the end of the day, it is faster just to do it manually (Yu et al., 2025).

All these different aspects can contribute to the cognitive overload of designers when using AI tools with no previous experience. Synthesised as “the mental resources demanded by a particular task”, cognitive capabilities can be overwhelmed when people are required to learn a tool, troubleshoot it, and interpret, analyse, and fix its outputs over and over again. Visual cognitive overload can in turn weaken the human brain’s processing capability and result in a worse performance. It is relevant to point out that the opposite can also be true: that automation can indeed reduce mental load, but the factors that influence the outcome cannot always be predicted and require careful analysis (Yu et al., 2025; Zhang et al., 2021).

In their study, Zhang et al. (2021) found that AI can actually inhibit high-performance team results while bringing some initial improvements to low-performance ones. The study involved different teams separated by performance (high, middle, and low) and AI assistance (hybrid or human only) with the goal of designing bridges. The algorithm used offered suggestions on the

bridge design through a heatmap where changes were highlighted in different colours. Although it seemed like the tool benefited decision-making, after the initial problem changed, the technology was rendered useless, with no improvements or deterioration. Interestingly, both middle and high-performance hybrid teams performed worse than the human-only ones while believing they did better. The misconception came from the false sense of success that the algorithm gave the hybrid teams that influenced them to become lazy and put in less effort into solving the problem. In other words, having AI at their disposal made teams rely less on their creative and problem-solving abilities and more on the tool that was actively hindering their performance. It made them less curious to try.

Other studies have found similar results, with AI-generated images functioning as a block for the engagement and motivation of brainstorming teams, and the user experience leaving underwhelming results (Yu et al., 2025). When Zhang et al. (2023) presented architects with a generative image algorithm that could create polished illustrations based on sketches, some of the participants really battled with the technology. Not only did the model not understand the technicalities that it was being asked to do, but the final images were also either nonsensical, far from what the architects had envisioned, or not up to standards to be shown directly to real clients. The general consensus was pretty divided, but while some took advantage of the tool to develop vague ideas, those who had specific images in their mind felt like they were fighting for creative power with the algorithm.

We can conclude that automation can directly impact intrinsically human qualities that are necessary for a successful AI adoption, such as adaptability, curiosity, and empathy (Bowles & O'Hanlon, 2025). In creative contexts, the loss of the human essence can be critical and pull designers back from their full potential. With that being said, how we define the line between use and over-reliance of algorithms is unclear and will depend on the context and environment of each project. In general, this multitude of challenges and risks brought by artificial intelligence can certainly feel catastrophic. From harmful fake news to surveillance and privacy issues, AI technologies are not perfect and will not be for a long while, but it is not a case of despair. There is an opportunity for every risk that will make the adoption of the technology worth it, if used correctly and with the right intentions.

2.4.2 Opportunities

Artificial intelligence generative models have become impossible to miss, and there are multiple reasons for their popularity. They are brand-new tools using an established technology that

many did not recognise, it is being deeply researched and invested in by the biggest organisations worldwide, and at the end of the day, it works. Machine learning and neural networks are huge drivers of innovation that are reimagining how we currently do things and actively creating value in dozens of industries and across hundreds of use cases, by applying numerous analysis and transformative techniques to vast amounts of data that is being collected in very short timeframes. Its capability to classify, cluster, optimise and generate new data are just some of the methods utilised by these models that make them so useful for corporations and individuals alike (Chui et al., 2018, Kaur & Sharma, 2021). From healthcare and sports to education and the automotive industry, there are countless opportunities to be taken. Keeping the scope of this study, the following benefits will be mostly focused on organisational behaviours, service development, customer and user experiences and the general design process.

The biggest advantage of AI is the possibility of automation of simple and complex tasks, and how it can streamline a lot of processes within companies. In other words, algorithms can take over certain low-skill, time-consuming tasks or sections of processes that would take humans a considerable amount of time and effort and not only do it faster but with more precision. The same can be said for some high-effort tasks that are being slowly optimised; AI can seemingly do it better. Automating the right tasks can free workers from overly repetitive and tiring tasks and allocate their skilled labour to more important functions that call for human know-how and emotional abilities. People who are highly adaptable and showcase a variety of competencies are key assets in companies that are looking to redesign jobs and optimise their employees' responsibilities through the replacement of unneeded positions, effectively increasing production and boosting economic growth and prosperity (Al-Kfairy, 2025; Chui et al., 2018; Kaur & Sharma, 2021). Although some jobs will most likely cease to exist, new AI-related jobs will start to appear in the future, with more than half of the children currently in school growing up to work in completely novel roles, with skills that are lacking in the current talent pool. (Santana, & Díaz-Fernández, 2023).

Most importantly, AI can be used side by side with humans as a co-author, helping in research, ideation, development, and prototyping. By making an almost infinite amount of information quickly accessible and synthesised, AI models give creators the opportunity to research, expand, and iterate their ideas, as well as create new things and discover new pathways that were previously unthinkable. Moreover, AI's analytical proficiency highlights specific data insights such as customer behaviour and employees' mood that essentially aids designers and users to

make informed decisions and evaluate the best course of action. (Al-Kfairy, 2025; Chanchamnan et al., 2022; Kaur & Sharma, 2021).

Not every task can be automated or augmented with the current level of technology we have today. The literature emphasises specific areas that are more likely to benefit from the adoption of AI systems than others, one of them being talent acquisition. Companies can use AI to pinpoint missing competencies or roles within the organisation and consequently write and publish jobs online. The algorithms can then match and screen relevant candidates, even using chatbots and other platforms to conduct artificially generated interviews. The same thing can happen the other way around; jobseekers can turn to LLMs to review their curriculum and adjust them according to different job positions, write engaging motivation letters, and enhance their portfolios. This can be particularly useful for designers to match their skills to relevant roles, and the creative industry often presents hiring advertisements that are not titled accurately or ask for additional skills that are not related to the job they are offering. For example, a graphic designer that does motion graphics and packaging, or a UX/UI designer that also does frontend development. Although it is not unheard of, and design is a multidisciplinary field, it can be hard to find the right role.

With the help of AI, designers can save a lot of time when applying for jobs or choose to add new skills to their repertoire, like learning the basics of code with Claude. After the hiring process, algorithms can also be used for employee retention and development, from the initial onboarding education to smoothing the employees' transition into their new role and environment, to continuous training over their careers that helps them improve their skills and keep their knowledge up to date with market expectations. This is especially suitable to address common job replacement and concerns over feeling left behind that lead to AI anxiety, by pushing workers to take part in technological innovation and learn about new tools by using AI-powered e-learning platforms. (Chui et al., 2018; Kaur & Sharma, 2021).

Another big use case of neural networks is risk assessment and management. Deep learning algorithms are able to detect existing anomalies in hardware and predict future failures. They can then use this information to forecast their maintenance needs instead of waiting for them to shut down. In a business setting, this helps companies reduce costs brought by the downtime of their own products by alerting the need for intervention. Translated to a service context, algorithms can highlight recurrent pain points in the user journey that would call for an update

in the experience (Chui et al., 2018). The financial department can greatly benefit from automated risk management by having it analyse the firm's revenue and spending trends and anticipate how a new change could affect their numbers. Together with the marketing and sales teams, AI can organise the launch of a new product or service, along with its advertisement campaign and guide the general decision-making in terms of budget, resources, audience, and timing.

The ability to personalise platforms, predict market trends, and understand consumer behaviour makes marketing, sales, and the public sector areas that can gain a lot by designing with AI augmentation in mind (Al-Kfairy, 2025; Mortati & Freitas, 2026; Singla et al., 2025). This technology can fundamentally change the way that organisations interact with their customers, improve decision-making, automate internal processes, and create intelligent augmented products and services (Marr, 2019). From the start, deep learning models can actively assist lead generation by helping companies find the right target audience that matches their services or switch strategies and uncover brand-new user segments that align better with the brand's goals. At the other end, chatbots and robocalls have already been implemented in customer service during the past couple of years, answering frequently asked questions and finding information for users while leaving human agents for more complex situations (Chui et al., 2018; Mortati & Freitas, 2026). Between these two touchpoints is the most interesting part, the process of co-designing a service with AI technology.

We can understand service design as having two environments and states. The backstage is where the strategy leads the process, the potential stage. It is here where resources are connected the service delivery planned, and the service designers shape and conceive the service encounter by aligning its value concept with its internal architecture, as in stakeholders, actions, touchpoints, and workflows. In the frontstage, the kinetic state, the service is actively co-created through the interaction of the users and the service providers. Without one or the other, the service would not exist. When we assume machine learning as a tool capable of creating value and co-creating the service experience beyond just automating tasks, we discover that the relationship between human and machine can take various forms in the service space. In Service AI, a hybrid environment, humans and AI act both as users and providers and work together across the back- and frontstage to co-pilot the service system (Mortati & Freitas, 2026).

Mortati & Freitas (2026) establish four types of possible interactions between humans and machines in a service context: human-to-human, AI-to-human, AI-to-AI, and human-to-AI. Neural networks are present in all the interactions, but their deployment depends on the nature of the service and the need for the technology. In a human-to-human setting, AI remains backstage where it uses data analysis to assist decision-making while humans carry out the service delivery. Like this, the traditional service journey, with personal social interactions, can be preserved while being quietly augmented by AI. This type of augmentation can be used to transcribe meetings, quickly analyse customer information and previous data, and offer recommendations in real-time, customised to the situation.

In human-to-AI, the machine takes the place of the user, becoming a non-human actor with unique needs and goals. In this scenario, the human-run service must convey human logic through formats that the machines can read and understand, like datasets and logical frameworks, information that is carefully structured to help train the AI. In business settings, this is present in neural networks that can autonomously maintain stock levels by placing deliveries and scheduling appointments directly with customers, effectively forecasting demand and supply and relieving humans from repetitive tasks. It is here where chatbots come in, algorithms that provide users with needed information through a human-like dialogue, acting as autonomous customer service agents.

The opposite happens in AI-to-human, with machines playing the role of service providers and the humans interacting with the algorithms instead. Whether it works alone or side by side with human collaborators, AI as a provider can automatically cater to specific users using data analysis and adjust the service interface accordingly. In the backstage, these services are co-piloted by designers who set the ground rules for the AI's behaviour and curate the outputs it generates through a cycle of prompting and iteration. This interaction can result in highly personalised platforms and interfaces that shape the service to each unique customer's experience, like streaming platforms Netflix and Spotify that already use algorithms to this end, offering customised content and recommendations based on the user's data.

Lastly, in AI-to-AI encounters, humans remain backstage as overseers of the service interaction. Considering that machines lack the need for social relations, this encounter consists of extensive data being communicated between agents, which is organised and evaluated by the humans responsible. Like in human-to-AI, the resulting outcomes are protocols, rules, and frameworks that are comprehensible for both machine learning agents and allow them to create

value in people's lives. The AIs can forecast users' needs and place orders, make grocery shopping lists, or even analyse traffic patterns to prevent congestion.

Across these four situations, we can see how humans remain in the loop alongside technology and how artificial intelligence can elevate the value that humans bring into services, business, and design processes. Although the machines can, in fact, automate workflows and improve experiences, it is up to the designers to carefully evaluate where and when AI should be deployed, and also recognise when it should not be used. It is also the designer's job to find alternative ways of using this technology that take it beyond data analysis. As previously mentioned, neural networks have the ability to recognise behaviour patterns and cluster similar groups of information that help designers define their target users. In the same way, it can understand preferences and needs, these models can also simulate real people by creating realistic and interactive personas for user experience (UX) testing. Although it can seem strange at first, following this technique can be extremely timesaving as it lets designers conduct user interviews at any time and place, in contrast with in-person observation and meetings that can be difficult to schedule and hard to synthesise the gathered feedback (Reznik, et al., 2026; Chanchamnan et al., 2022).

Contextual awareness and emotional intelligence are some of the human characteristics that deep learning models still struggle to perfect and benefit from having a human to guide them to understand how they should be applied. In user experience design, similar to customer experience (CX), empathy is a key asset that can be augmented by AI. Even with current studies defending how LLMs can successfully identify emotions in different situations and imagine action paths according to other people's feelings, essentially showing empathy towards others (Schlegel et al., 2025), there is still a gap between this finding and how it can be used in real cases.

Connecting with users' emotions is a big step towards ensuring they have a pleasant experience interacting with products and services in their day-to-day lives. Integrating emotional intelligence principles into neural networks could address common issues such as bias and lack of transparency and enhance human-centred design by aligning goals and needs. While fully emotional AI is still in development, its ability to recognise and process body movements, facial expressions, and tone of voice can help people identify easy-to-miss social cues in certain situations like online discourse; offer real-time coaching to people in their own EI journey by

encouraging self-reflection and suggesting relevant resources; adapt messages or email responses according to the user's tone of writing; and augment regular decision-making by considering emotional aspects alongside traditional data (Singh et al., 2024).

With how quickly the technology industry is advancing, some researchers predict that it will not be long until emotional intelligent robots and machines are available to the public and present a huge opportunity area for user experience designers. Using facial recognition, services will be able to measure customers' emotions towards different aspects of the user journey and use those insights to consider prices, organisation, advertisement, and product development. An emotional AI agent could take advantage of this data to offer personalised recommendations and support (Kaur & Sharma, 2021). That being said, it is pivotal to adopt ethical data collecting methods in a way that respects the public's right to privacy and data security, which could also be ensured by context and empathy-aware AI models.

Machine learning is getting more human-like with every iteration cycle, learning how to be empathetic and mimic human creativity. Although it may feel uncanny, the evolution of technology from data-processing rational algorithms to complex agents that openly communicate brings new potential to the creative industry. Image generators and LLMs are a democratising force in the art environment, providing intuitive tools that everyone can use, no matter their age, background, health situation, or artistic ability. Although art is a universal language and has a relatively low entry barrier, everyone is now capable of creating high-resolution artefacts to transmit their ideas and feelings, which can be more impactful than simple low-fidelity results. Those who feel self-conscious about a lack of skill or creativity can use AI technology to help them gain confidence in their creative abilities and use tools to help them practice real, tangible methods. Professional artists who specialise in specific techniques like painting and illustration can take advantage of these tools as well, as a means to explore alternative mediums like music, animation, or graphic design (Garcia, 2025; Wu et al., 2021). It must be noted that these tools should be used to supplement unique concepts and deliberate artistic intentions and not as a replacement for human imagination for the purpose of just making something. At the same time, it is beneficial to explore the limits and capabilities of generative tools and have fun using them, but it is not advised to do so in a professional artistic setting.

AI can bring down barriers and encourage an exchange of different cultural influences. Artists all over the world can dissect and combine their works with references they would have never

thought about and project them into channels they would not have access to, for example, creating a poem, painting, and song based on a language they do not know or imagining a huge mural on the side of a building (Garcia, 2025; Wu et al., 2021). Co-creating with AI supports analogue thinking, also known as lateral or divergent thinking, the process of breaking down and translating ideas by finding similarities between different concepts. Like metaphors, designers use known relationships between things and apply them to different sets of things, fostering unexpected results. One of the most relevant benefits of this collaboration is the effective combat of creative block or design fixation in the creative process. In other words, when creators become fatigued and overwhelmed with the ideation process, it is common to become fixated on one or two ideas and not be able to think outside the box. Traditional methods to mitigate this phenomenon include taking a break and asking for feedback from an outside source, which can be both frustrating and time-consuming. Generative models can be a great inspiration source, proposing a vast number of alternative concepts that stimulate lateral thinking and allow designers to reach innovative ideas. Simultaneously, the cognitive aid given by AI tools can lessen the mental load of designers in the ideation process, by helping them quickly define a strong concept rather than struggling with loose ideas for days (Al-Kfairy, 2025; Lee et al., 2025; Mortati & Freitas, 2026; Yu, 2025; Zhang et al., 2021).

Not everything that designers do is related to actual design work; a lot of needed tasks are considered boring, including writing emails, organising documents, and adjusting minute details. Freeing humans from repetitive tasks like spam filtering and writing reports allows designers to focus their skills on more complex and strategic creative work instead of mundane tasks that often get in the way. Machine learning models excel at repetitive types of tasks and can step in to create cohesive and human-like communication materials, routine reports, and design social media posts that adhere to each platform's style, from LinkedIn’s casual professionalism to Instagram’s playfulness. On top of that, it can optimise team communication through automated project updates, alerting staff about policy changes, and sending company-wide reminders for important meetings and deadlines. AI can also be a powerful evaluator of internal processes and creative artefacts besides just task automation. It can make sure design materials follow the brand’s guidelines, avoid possible restrictions, and always maintain the tone of the company’s identity (Al-Kfairy, 2025; Chanchamnan et al., 2022; Yu, 2025).

There are multiple layers of co-authorship that AI can act on. On a business scale, it can reshape how customers interact with services, foster productivity, increase support for employees in

their careers and wellbeing, and help firms innovate. We have also observed how team communication can be optimised, making sure everyone is on the same page and has access to the same tools, fostering proactivity from each individual. Perhaps the most interesting aspect is how this technology collaborates with humans on the individual level.

As previously emphasised, AI brings incredible amounts of data directly to the hands of the public, synthesising important information and mixing different influences that push humans to view things through different perspectives. Brainstorming becomes playful when exchanging ideas with machines helps us imagine new worlds of possibilities, allowing us to learn together, make sense of the world through accessible tools and language (Dwivedi et al., 2021), and perceive a reality where AI adapts to the human context. This is when AI becomes a part of our lives as "ambient intelligence" (Wu et al., 2021, p. 172) spaces that adapt to the human presence and co-create experiences with us (Wu et al., 2021).

Beyond cognitive activities, neural networks facilitate creative expression. Images, videos, audio, and text can be generated to support a variety of projects. Musicals? Video game assets? Websites? Personas? Everything can be generated one way or another, being in small pieces or as a whole artefact, based on artists' drafts or functioning as a stepping stone for the development of art. They allow ideas to evolve in directions previously unknown and give artists a sense of liberty to try many different things in a bigger design space. Tools that turn sketches into concept designs allow designers to create high-quality images much faster than doing them by themselves, giving them the time to generate double or triple the variations they could before. From the beginning, artists can brainstorm with machines that give them insightful inspiration, co-create a strong concept, and engage generative tools to visualise said ideas with the precision of final concepts. The freedom of this process lets artists iterate their own ideas tailored for different requirements and swiftly adapt them depending on feedback from clients or collaborators (Al-Kfairy, 2025; Chanchamnan et al., 2022; Yu, 2025; Zhang et al., 2023).

The ease with which deep learning methods can recommend alternative ways to represent a design in different mediums and formats makes AI a great guide for cohesiveness in a creative project. It can give insights into how elements can be shown in different spaces and make changes based on pattern recognition and analysis. In a UX scenario, being able to customise artefacts at scale is an incredible time-saver for designers and it results in products that are better aligned with the user's needs. Moreover, machine learning is capable of analysing user testing videos, workshops, and interviews to automate feedback that can validate the designers'

own findings. Holistic feedback is, overall, a huge benefit of using AI in the creative process. Similarly to how it aids brainstorming and ideation by collecting different data, it can use the same mechanism to provide clear feedback notes and help creators understand how to move forward. In the prototyping stage, this means rapid evaluation that allows products and services to be thoroughly tested and improved over time, producing easily adaptable transformative design solutions. (Al-Kfairy, 2025; Chanchamnan et al., 2022; Mortati & Freitas, 2026; Yu, 2025).

2.5 Human-centred AI adoption strategies

2.5.1 Requirements

Having contextualised both the risks and opportunities that machine learning brings to companies and the design space, we can now turn to the implications that come with the ethical adoption of this technology. Before delving into strategies, there are ideals that organisations must adhere to for a successful AI integration within their teams and leadership. It is essential that businesses strive to avoid the ethical issues associated with AI systems that may impact their employees and consequently negatively affect their performance and public reputation. Having an agile and ethical management style is the first step to take towards that goal, as they will be more likely to care about the wellbeing of employees, address the initial resistance to change, and be open to accepting alternative innovation methods within their company.

Besides a skilled management team, firms should work towards having AI experts within their teams, either through new acquisitions or internal training, to be able to maintain the systems in place and also be acutely aware of the mistakes and the common problems resulting from AI tools. It is important that staff have an understanding of how the technology works and what they can do in case of failure or shutdown and be critical of the outputs it can generate.

Data security is one of the main aspects that influences the amount of trust stakeholders put in a company, and it requires a constant evaluation of quality throughout the whole process, from before to after the implementation. It is also necessary to have enough data before considering the transition to neural networks, that is collected ethically. Both the quantity and quality of the data are crucial for AI systems, as they shape the performance of all the models that are trained off of them, inheriting any bias and misinformation that may be present. For this, firms have to have governance guidelines that ensure the privacy of the owners of the data, like laws that restrict the usage of the data without explicit permission or for nefarious purposes. The better

the security around data, the more confident stakeholders will be while using the new systems and the more open they will be to accept innovation (Al-Kfairy, 2025; Merhi, 2023).

Because of the processing power needed to collect copious amounts of necessary data, the implementation of AI requires a stable IT infrastructure that is up to date and able to undergo considerable updates to host brand new systems and data storage. Investing in new technological resources comes with the need for continuous updates and a demand for a grand financial capacity that includes enough budget to cover any associated expenses. The new AI systems will need to be maintained, understood by internal and external experts, and upgraded as the technology evolves, calling for adaptable, scalable systems to be in place (Al-Kfairy, 2025; Merhi, 2023).

In SMEs, these requirements are even more emphasised due to limited resources. With shorter budgets, a lack of external financial support, and smaller teams, firms of these sizes need to focus their goals on obtaining affordable solutions with a guaranteed return on investment. This includes starting with smaller systems that can grow along with the company and finding collaborations with key enterprises that allow for flexible payment methods or service exchanges. Similar to bigger enterprises, SMEs need to create AI literacy within their company. Instead of outsourcing or new acquisitions, businesses should be responsible for training their in-house workers and generating internal knowledge (Al-Kfairy, 2025).

Regardless of scale, it is imperative that all businesses assume an open-minded, agile governance model that prioritises fairness, accountability, human rights, transparent data collection and use, and strive for long-term ecological sustainability. It is also crucial that all actors involved are aware of the risks and opportunities that AI can bring to an organisational setting and are ready to manage eventual financial, infrastructure, or interpersonal pushbacks that may happen.

2.5.2 Strategies in Literature

Enterprises have started to adopt AI across the board, especially in the technology, media, telecommunications, and healthcare industries. During a highly experimental time, firms are currently deploying pilot projects with AI agents, testing which has shown a significant impact on innovation growth, customer satisfaction, and competitive advantage. While earnings have not yet been affected at a grander scale, some cost decreases have been shown in IT and manufacturing companies and bigger benefits in marketing and sales, as well as product and service

development. From a broad perspective, the leading businesses in AI implementation showcase committed leaders focused on strategic initiatives that foster transformative change while abiding by ethical practices and choosing the right investments. One of the biggest differentiation factors is the attitude with which firms face this technological shift, where those who embrace ambition, create strict ethical guidelines, have a set vision for the future, and a defined strategic roadmap to reach it, find highly successful results (Singla et al., 2025).

Corporate social responsibility (CSR) is a business model common in organisations with agile governance frameworks, where companies manage their impact on society, environment, and stakeholders with ethical concerns over workers' rights, sustainability, accountability, and data. With the aim of effectively collaborating with technology rather than fighting it (Wu et al., 2021), corporations apply strict standards, ethical AI toolkits, and legislative rules to mitigate recurrent issues like data security breaches and extensive layoffs. Data governance models are established, and there is a strong focus on upkeep and managing systems over time (Chui et al., 2018). It is worth noting that these guidelines are not enforced by third parties, and it is up to each company to promote their own responsibilities instead of partially committing to values whenever they see fit or using them as a PR tactic and not an honest endeavour (Bughin, 2025). Including all stakeholders in the process of creating these frameworks, including specialised lawmakers and employees from various sectors, allows the resulting strategies to represent diverse perspectives and not benefit the agendas of specific powerful individuals. Risk management greatly benefits from a participatory approach where employees are encouraged to identify ethical and bias concerns they come across, supporting a collaborative evaluation and subsequent co-creation of mitigation actions and implementation strategies that favour as many stakeholders as possible. To facilitate the mediation from the staff and various actors, effective communication channels must be implemented for feedback and improvement suggestions (Bowle & O'Hanlon, 2025; Madanchian & Taherdoost, 2025).

To ensure the constant monitoring and review of the process, from risk identification to the creation of strategies, as well as the confirmation of best practices, companies should consider establishing an internal AI ethics committee (Bughin, 2025). The goal is to consider every possible risk, no matter how small, and be prepared to act swiftly in case anything happens (Makridakis, 2017). This group of experts would be responsible for advising the best courses of action, including the implementation of emerging solutions to foster competitive advantage and securing the trustworthiness of the company both internally and with the public. Users have the right to know how much AI was used in the creative process and contributed to the design

of a product or service, which can be transmitted through a label (Garcia, 2025), and employees deserve a Taylorism-free environment, where data surveillance is not used against them and the goal is not to push efficiency to the maximum (Bowles & O'Hanlon, 2025). Like this, organisations can guarantee that their customers choose their services out of choice and loyalty and not through dishonest methods and that their workforce is proud to work for the company and is not unnecessarily overworked. With good customer service, happy employees, and a strong competitiveness in the market, businesses are more likely to remain relevant in a sea of innovative startups and AI-powered firms (Makridakis, 2017).

Collaboration continues to be the key to a healthy working environment that prioritises people-centred perspectives. Instead of maximising profits and automating whole processes for the sake of efficiency, maybe the solution is to use technology to help people grow while maintaining necessary social interactions and creative outlets that keep humans sane. The reality is that people need things to do, hobbies, tasks, responsibilities, because it gives them purpose and a reason to learn and experience the world. Using AI with empathy can make this the case for everyone, reducing stress on laborious tasks and connecting people to roles they enjoy having (Jak Spencer as cited in Dwivedi et al., 2021).

Approaching AI with a mindset where technology can handle "data-driven decision-making" and let the strategy and conceptual thinking for humans (Yu et al., 2025, p.638) allows human value to be amplified (Dwivedi et al., 2021; Santana, & Díaz-Fernández, 2023) and prevent AI slop. The misuse of generative programs to quickly produce content or with little to no oversight (Resnik & Hosseini, 2026) not only undermines the worth of human nuanced judgement and imagination, but it also weakens organisations' reputation and trustworthiness (Dwivedi et al., 2021). Machine learning cannot achieve the same quality and strength of concept as designers can and should not be used to replace the final stages of creative work to preserve designers' mental health and ensure that projects and communication translate the high standards of each company. Artists like doing art and will understandably refuse to accept a technology that will prevent them from doing their craft (Garcia, 2025). The same way, designers will abdicate from using AI if it is brought as a replacement for the interesting part of their work, pushing them to the side as a corrector.

It is crucial that each firm invests in mandatory training on responsible AI use for all employees, including general best practices and specific requirements for each role (Bughin, 2025).

Curated learning tools, as well as effective recruitment and onboarding flows, guide the workforce through professional development in the current context and facilitate the acceptance of the evolving changes (Singla et al., 2025). As technological industries continue to evolve, robotics and machine learning are working towards a future where robots will be able to move and operate similarly to humans. The importance of applying AI to foster long-term human success is increasingly clear when we realise that most jobseekers do not have the necessary skills for the industry these technologies are shaping. As a result, competencies like critical thinking and being able to adapt to new situations become increasingly useful in society to enable the redesign of traditional roles (Santana & Díaz-Fernández, 2023). When paired with collaborative systems thinking that aligns technical capabilities with human-centric objectives, these frameworks prove that human-centred strategies are the true drivers of successful and sustainable AI adoption journeys. Beyond workforce training, it is necessary to create adaptable leaders who are open to exploring alternative methods and can confidently guide their teams through unpredictable socio-technical shifts (Bowle & O'Hanlon, 2025).

The Human Capability Standards (HCS), created by The Institute for Working (as cited in Bowle & O'Hanlon, 2025), offers a framework that puts non-technical skills on the spotlight while addressing anxieties over job loss. Instead of specific practical tasks, HCS emphasises abilities that can be transferred between jobs and roles, freeing the workforce from strict professional roles. As the job market undergoes major transformation and current positions lose their relevance, AI can be applied as a learning tool that builds careers and boosts job quality through the development of digital competencies (Bowle & O'Hanlon, 2025). It must be recognised that technology-related competencies are not the only ones relevant to support AI adoption; this journey implies multidisciplinary knowledge as well as social and personal soft skills. Being independent and proactive, having cultural awareness and empathy, knowing how to learn and translate information into expertise in different areas, understanding online behaviour including having a high level of media literacy, content creation, data sharing and management, and digital safety, and of course, critical thinking and problem solving, are just some of the competencies that are currently on top of the value chain (Santana, & Díaz-Fernández, 2023).

Hiring high-quality skilled experts with new competencies is a big step towards ensuring that there is a strong level of knowledge to confidently handle AI and foster an innovative environment (Makridakis, 2017). Technology-related workers are the ones who will be in close contact

with the technology and manage any unfortunate situations that may arise, and non-technological ones will oversee different disciplines affected by AI, like the HR, finance, and legal departments (Singla et al., 2025). Having a workforce plan is particularly important if companies choose to develop their own AI systems, as they will need to attract and keep valuable human resources to implement and upgrade the networks (Chui et al., 2018). Creating an independent system permits firms to not be reliant on foreign technology and to apply ethical guardrails from the beginning of the process, having control over data collection and processing, preventing bias. It would also assure data security and privacy. However, a project like this needs an incredible amount of funds, time, and infrastructure that not many companies have access to. Another option to acquire specialised workers is to collaborate with universities and other educational institutions, as students are not often tied to workflows, are excited to explore, and have insights into the latest trends in the industry (Santana, & Díaz-Fernández, 2023).

A culturally aware staff can help mitigate issues like LLM hallucinations. When around one-fourth of LLMs' content is hallucinated (Ahmad et al., 2023), it is mandatory to have evaluation processes in place. It can be as simple as asking the model how it came up with its answer, comparing results with the data source, and being careful not to sample a generated response multiple times to avoid contradictions. To address the issue from the source, there are multiple mitigation strategies that can be used in tandem. Starting by choosing trustworthy AI platforms that are proven to demonstrate fewer mistakes, using a human-in-the-loop approach to analyse the outputs, suggest corrections, and supervise the machine's performance in real time if needed, prompting the LLM to admit when it does not know something and "punishing" it when it lies or omits the truth. If the data comes directly from a company, validating the input data can also be very helpful to flag down any mistakes or bias (Ahmad et al., 2023; Chui et al., 2018). Keeping humans in the loop makes sure that generated results are not taken at face value and are fact-checked by real people. This can cause trust concerns to arise as the workforce realises how much of the generated content is incorrect, to which the company should respond by reinforcing the ma-chine-human collaboration and reassuring its employees that it values transparency, trust, and humanity (Dwivedi et al., 2021).

Designing with AI involves developing clear limits for what should and should not be automated or augmented, and how intensely they will be. More specifically, design teams must select tasks that they consider can be improved by machine learning, debate if they are fit for automation or augmentation, choose the level of machine intervention, and elaborate strategies to manage output errors and the relationship between humans and neural networks. Moreover,

tasks can be separated into different stages that can integrate deep learning in different ways, or maybe not at all. It is up to each team to consider how useful automation would be to different sections of designers' workflows to avoid unnecessary changes with no explicit objective (Dwivedi et al., 2021). With a good task allocation plan, design teams can optimise their work and have more time to explore designers' cognitive skills. By having generative models assist in brainstorming and visualising concepts, creatives take on the new role of "Designer Arbiter" (Figoli et al. as cited in Yu, 2025). In simple terms, designers gain the responsibility of critically analysing AI outputs and evaluating how they can be implemented in projects and processes. Like this, designers preserve their agency by using their creativity, intuition, and emotional sensibilities to dictate how technology is used according to their needs (Yu, 2025). The redesign of workflows and goal-oriented use of deep learning models also serve as a method to manage "technostress", the anxiety that comes with using new technology fuelled by the overwhelming number of models and possibilities available (Reznik et al., 2026; Singla et al., 2025).

From hundreds of different established and personal frameworks, the double diamond method is one of the most recognised workflows in design. Both simple and memorable, this model covers research and brainstorming in the Discover phase, concept and ideation in the Define phase, creation and iteration in the Develop phase, and testing and prototyping in the Deliver phase. Although different, other frameworks tend to follow a similar structure that includes these main stages of the design process.

Large language models and generative programs can bring multiple benefits to the design process. In the beginning stage, LLMs give designers access to large amounts of data that can be organised to reveal user trends, usability needs, and uncover niche issues. Designers can use these models to explain and visualise complex problem spaces, explore interconnected systems, and understand design principles before starting a project. The technology allows for a high level of customisation that can be used to create graphs, roadmaps, mind maps, and even host brainstorming sessions, essentially acting as a design assistant. Moreover, it is at this stage that creatives can consider a variety of different concepts and discover different pathways. Machine learning can highlight real projects and mediums that would otherwise go unnoticed to designers. Unknown references from around the world, books, movies, and paintings serve as unexpected inspirations to create more unique concepts while helping teams combat fixation and maintain a smooth creative flow (Lee et al., 2025; Yu, 2025).

After the initial information overload, the same tools can synthesise the information into simpler terms using qualitative coding that helps designers define the design problem and the purpose of the project that will address it. This means translating vague ideas and mood board pictures into concrete concepts, offering virtual spaces for experimental ideation and experimentation, using sketches to create vectorised UI wireframes, and overall supporting the choice of design themes. Furthermore, deep learning models can delineate design constraints so that creatives spend less time in trial-and-error approaches to understand what they can or cannot do with the data and materials they have. To better immerse themselves in the data and get closer to their target, designers may use chatbots to create user and client personas based on the collected information that can interact with the design team to express their desires and complaints directly (Lee et al., 2025).

In the developing phase, AI is at its most useful as an aid for the development of solutions. In other terms, different tools can be generated and used to facilitate processes and make it easier for designers to reach their desired results. Programs that generate grid layouts for digital wireframes and organise information hierarchy for websites, seamlessly apply a design style across different digital products, generate shapes and audio descriptions, or allow users and designers to provide immediate feedback on generated assets are just some of the possibilities that AI brings to the industry. With enough knowledge to programme a personalised tool that addresses specific needs for each project and team, designers can create their own platforms to assist decision-making and problem-solving. Being able to upload their first sketches and ideas into a generative model that turns them into complex illustrations or 3D models not only grants creatives the opportunity to quickly visualise their ideas and add on to their concepts, but it also makes them interactable. Designers can then explore virtual spaces and products with changeable components and easily generate changes and variations of the same concept. Prototypes become more real and adaptable to feedback. Because of the iterative nature of the developing phase, it is the section with the most potential for AI augmentation through automated code generation, simulation and analysis of user data and behaviour, and data-driven refinement of ideas (Lee et al., 2025; Yu, 2025).

Lastly, before the final delivery of the project, design teams can assess the feasibility of the solutions developed. To that end, designers can not only collect feedback from user testing, but they can also generate their own feedback with chatbots and LLMs. Generated feedback that is fed constraints and user data can be critical of designs that go against those requirements and suggest alterations that better fit the objectives of the work. At the same time, these tools can

effectively analyse external feedback and compile important insights of what is working well and what needs to change. High-fidelity prototypes are also a possibility and can be explored together with generated users to observe how humans would interact with products and spaces. This type of virtual user testing reduces costs and time constraints of doing it with real people, although it should be carefully considered and complemented by real human feedback.

The same way a hybrid double diamond can be used in UX, UI, and product design, illustration, and branding, it is also applicable in the service. In that context, designers must deliberately decide when AI will be present and visible to the public and find unique ways to represent the technology using prototyping tools. Methods like user journeys and service blueprints must become dynamic to be able to mimic the fluidity of AI technologies like chatbots and LLMs. Especially in cases where AI takes on the role of a service provider, workflows need to maintain its interactivity post-launch, to keep up with the evolution of the machines and be able to guide them on the right path. With the lack of established guidelines due to the novelty of the current artificial intelligence programs, it is up to design teams to test their own hypotheses and complement the double diamond to fit their needs. (Mortati & Freitas, 2026)

It is important to note that, as mentioned previously, any and all use cases of AI should undergo a strict human-centred and informed decision process alongside the designers to reach a workflow that works for everyone and keeps the creatives engaged in their work (Singh et al., 2024). Like this, the technology also works as a support for team communication and collaboration, taking a mixed-initiative co-creativity (MI-CC) (Yu, 2025, p.11) approach. That is, an environment where both humans and machines contribute to the creative process, with AI participating dynamically to offer suggestions, insights, and feedback as designers work. For companies that are interested in developing an internal model tailored to their design needs, a competence-centric approach is recommended to ensure designers work side by side with engineers to create a system that is not dominated by the designers and forces the technology to fit in or by the AI model and forces the designers to follow its requirements. By focusing on competencies and not having the model as the core of the project, they can guarantee the process was collaborative and the model is a result of a symbiosis between design and technology (Yu, 2025).

Finally, it is relevant to highlight strategies for small and medium enterprises. Sánchez, Calderón and Herrera (2025) explain how the Technology, Organisation and Environment (TOE)

framework, that looks at AI adoption through three dimensions: Technological readiness, Organisational capacity and Environmental conditions, and the Diffusion of Innovations (DOI) theory, that examines how innovations are perceived by the general public and organisations, can be used together by SMEs to evaluate their strategies for AI adoption. While the TOE framework assesses the infrastructure, financial capabilities, workforce skills, leadership, external policies and the very important ethical guidelines, DOI adds the human layer that often drives decision-making in smaller companies. It is with DOI that individuals evaluate the advantages of new technologies and ponder over its usability, compatibility and compatibility with existing systems and workflows (Sánchez et al., 2025).

To complement all previous strategic recommendations tailored to SMEs, there are additional initiatives that these institutions can take into account. Regarding lack of skilled labour, beyond collaborations with universities and internal training, firms are encouraged to participate in open innovation networks to exchange knowledge and gain exposure in the industry, by creating alliances with various companies, technology providers and leading firms. Aiming to maintain agility, it is ideal to choose modular and scalable approaches, by starting with smaller AI projects that address specific needs within the company and using cloud-based infrastructure that does not require complex hardware systems or a lot of power and storage. This also allows for future integration with new technologies and keeps companies' freedom to adapt and iterate their IT systems. This means that enterprises can start with simple and highly tested tools in the market and slowly start implementing them in more and more operations.

For example, the use of Gen-AI can start simply by drafting documents and then transition to automating routine analysis and responding to customer emails until it is fully embedded in the company's workflows and platforms. Moreover, there are opportunities for these institutions to use loans and public grants to their advantage, as well as collaborations with agencies and external hubs that give them the financial stability and expertise to start implementing pilot projects before committing to full-scale ones. To address augmented hybrid team organisation, the centaur and cyborg metaphors offer some solutions. The centaur model suggests pairings that perform tasks together that result in better outcomes than a human or machine would produce individually, and the cyborg represents workers that embed AI into their tasks to augment their performance in real-time. These are interesting approaches that are valuable to explore in initial pilot tests until companies understand which one works best for them or depending on different projects (Sánchez et al., 2025).

With that, the proposed roadmap for SMEs consists of six phases: assessment of current state; definition of strategic goals; selection of AI solutions; implementation of pilot project; management of training and scaling. As a general structure, it lets businesses complement it with their own empirical data and insights, while having a plan on where to go next.

As a first step, understanding the current situation of the company is essential to know where they need to improve: IT infrastructure, alignment in leadership and workforce, financial capabilities, and digital competencies. After analysing its weak and strong points, the firm can identify its objectives. Keeping their business goals and customers' experience in mind, business objectives can be defined, from small, easy-to-achieve ones to long-term strategic projects. In sequence to that, they can decide the best solution pathways for their organisations, considering their resources and needs. Depending on financial capacity and in-house technology knowledge, they can opt for off-the-shelf programmes or create custom ones internally. The implementation of the strategies should start small, with pilot projects that help assess the solutions before going full scale, as it also helps mitigate risks and create a trusting environment in the new changes. Along with implementing solutions, the company must offer constant training and support to their employees to ensure a smooth transition and prevent resistance. The goal should be augmenting the existing human value and following ethical best practices across the board. Lastly, as an iterative process, there needs to be an evaluation phase that measures established key performance indicators (KPIs) that can be used to enhance the implemented strategies. Depending on the results, companies can strive to scale their initiatives to the whole organisation or refine their pilot projects until they show higher promise (Sánchez et al., 2025).

3. Design Research Methodology (DRM)

This section describes the methodology used in this research, specifying the design research methodology (DRM) that guided its research, the subsequent method deployed for each phase of the process, and showcases its results.

The DRM framework, developed by Lucienne T.M. Blessing and Amaresh Chakrabarti (2009), aims to solve the lack of established guidelines and scientific rigor in design research. By offering a model with specific steps, this framework hopes to help researchers develop more effective and high-quality research projects that follow a solid line of argumentation. Due to the heuristic nature of design as a discipline, this method is highly flexible and can be adjusted

depending on different needs, wants, and requirements. The goal is to guide the research and not restrict it (Blessing & Chakrabarti, 2009).

For this end, there are four different stages that compose the DRM framework: Research Clarification (RC), Descriptive Study I, Prescriptive Study (PS) and Descriptive Study II (see figure 2). These do not need to follow a strictly linear timeline and often overlap one another during the process.

To begin the DRM method, one should find relevant literature that supports the focus and the goals of the project. The research clarification stage sets the path of the research by defining the questions and problems to be addressed, areas that need to be explored, and by developing an initial idea of the current situation and the ideal situation the designer is aiming for. In sequence, the Descriptive Study I has researchers analyse the data acquired from the literature and find which factors are crucial in the current situation to be defined in detail and studied in later stages. Thirdly, in the Prescriptive Study, researchers begin to ideate a new support to improve the existing situation. It is here where they describe how addressing the chosen factors will influence the present context and develop their ideas.

Finally, the Descriptive Study II's main objective is to evaluate the impact of the developed support and its success in achieving the desired situation, as well as identifying improvement opportunities for future iterations. This includes conducting multiple empirical studies, testing the resulting support in real scenarios (Blessing & Chakrabarti, 2009).

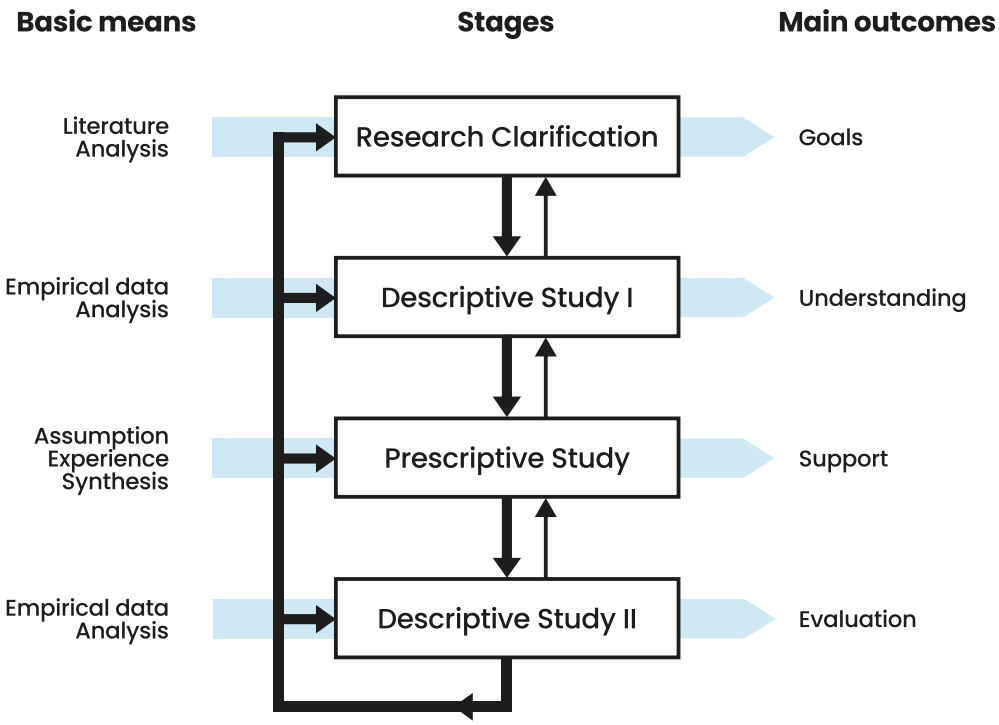


Figure 2 - DRM framework adapted from Blessing & Chakrabati (2009, p. 15).

In this research, I followed the type 2 of the DMR framework (see figure 3). My theory section in chapter 2 serves as the RC stage that serves to identify the current AI landscape in design, including prominent issues to be discussed and an overview of various approaches and opinions on the topic. The Descriptive and Prescriptive Study, where I categorise, analyse, and ideate solutions based on the empirical data I gathered, will be detailed in the following sections. This study does not offer a DS II and will conclude with an initial PS where the supports will be presented without further evaluation and iteration sessions.

Research Clarification	Descriptive Study I	Prescriptive Study	Descriptive Study II
Review-based →	Comprehensive →	Initial	

Figure 3 - DRM type 2 adapted from Blessing & Chakrabati (2009, p. 18).

3.1 Descriptive Study I

With the goal of answering the sub-question "How are designers currently collaborating with AI tools in their everyday work?" this section gathers qualitative data through a triangulation and ethnographic approach. It utilised multiple data sources and methods: interviews, conversations and non-participant observation where the author explored her targets' experiences first

hand, and a survey. The data collected from the literature in Chapter 2 was used to guide topics of interest and all methods of data collection.

For the purpose of this study, the design team, complemented by some web developers with design experience and tasks, was the main interest of the empirical research, as the goal of the research was to understand how AI can be used together with human creativity and agency (Blessing & Chakrabarti, 2009).

3.1.1 Non-participant Observation

Taking a fly-on-the-wall approach, observation was mainly conducted in the space of five weeks, three to five days per week, and between 11:00 and 17:00. The in-person observation happened in Liip's Lausanne office, with one day in the Fribourg office where a team was actively experimenting with AI in their workflow. The goal of this method was to blend in with the employees and maintain some distance to see them work and assess the environment of the company; therefore, it called for non-participation. This is, I did not interrupt or take people out of their work to answer questions or intrude in other ways.

Within the observation, I included the attendance of multiple "Liip talks", short presentations organised by Liip's AI changemaker (a role held by two Liipers who notably also fill the learning & development lead, production lead, and sales lead) where different team members showcased their current AI exploration projects in a show-and-tell model to foster collective understanding and motivate exploration. Some examples of these presentations included using Cursor and Claude AI to automate the coding process and programming skills (a set of instructions in Claude Code that teaches the programme how to handle specific, repeatable tasks) to automate specific steps or revise and refine a plan on its own. These demonstrations, although focused on programming and web development, revealed relevant insights into what was being done on an individual level and also some challenges people were facing. In total, I watched three Liip talks with a sum of five people presenting their experiences. Notes regarding these presentations, GenAIs and LLMs I saw people use, overheard conversations, and the general mood of the organisation were taken during this phase. Although it was focused during five weeks, this method was used throughout all my thesis as I continued to go to the office for a minimum of two days per week after that period.

3.1.2 Informal interviews

By informal interviews, I mean unstructured conversations based on relevant topics for this research and questions I believed would help me gather a general understanding of Liipers' mindset regarding AI use. The goal of this method was to get closer to the employees and understand their thoughts on the topic in a casual way, as in to let them feel comfortable to speak freely. Participants requested to remain anonymous with no direct quotes or mentions of names but agreed to let their data be used for this study and for internal research at Liip.

The conversations were mostly carried out during lunch hours and other breaks where employers felt comfortable and open to talk for 10 to 30 minutes each. I spoke with multiple people, from whom 12 showed relevant insights, and six of whom I had longer, more structured interviews where they showed me their work and told me about their processes. Some of these talks happened in the last phase of my research to offer some feedback on the support I developed and allow me to iterate my ideas.

The criteria for these conversations were not strict and started as a chance to speak with a lot of different people from different teams. These included web developers, analytics experts, programming apprentices, designers, strategists, and system administrators. After that, I scheduled proper meeting times to interview the six Liipers mentioned previously who fit in one or more of the following criteria:

- Were designers in any relevant field: UX/UI, graphic, service, web, etc.;
- Were working with different AI tools beyond programming, with design use cases;
- Presented relevant insights in the use, management, and future of AI use at Liip;
- Had relevant knowledge in the main pillars of AI adoption such as data collection and privacy, sustainability, or resistance.

At the end, the participants were a UX designer and web developer, a product owner, an AI changemaker, two senior designers, and a sustainable web designer. These interviews took place in person or online and took between 25 to 50 minutes.

3.1.3 Survey

Based on insights gathered from one-on-one conversations, a short survey of eighteen questions (including follow-up questions) was created to be sent to the general workforce at Liip. The

survey was created on Google Forms and subsequently reviewed and sent out by the AI change-maker. It was communicated clearly that the information was anonymous and the data would only be used for the purpose of this study and the AI changemakers' own research. No names were requested or mentioned, and no quotes were taken from the answers in the survey to maintain anonymity. The goal of this method was to gather more detailed data and answers, offer Liip complementary information for their internal strategic project, and reach a wider audience compared to the previous methods.

The survey questions went as followed:

- Role-type *(to be able to select answers based on role)*
- How do you expect AI-driven automation to influence your work in the next 3-5 years?
- Are you worried about the increasing use of AI in general?
  If yes, what specifically worries you?
- Do you feel pressured to use AI to avoid falling behind in your professional life?
  If yes, where is the pressure coming from?
- Do you want to use AI more?
  If yes, is it clear where you would start? If not, why not?
- What do you want to use AI for?
- Are you aware of the hallucination risks of LLMs?
  If yes, what steps do you take to prevent and/or deal with them?
- Do you feel like you have enough support on adopting new AI tools from Liip?
- How have the weekly AI Liip talks benefited you?
  (e.g., new tools discovered, inspired projects, improved technical understanding, or no clear benefit).
- If you currently use AI in your workflow, where do you use it the most?
- Do you have political concerns or boundaries in your usage of AI?
  If yes, what?

- Do you have ethical concerns or boundaries in your usage of AI?
  If yes, what?

These questions aimed to understand the current situation of AI use across different teams, especially designers, and what their approaches were. It included multiple topics from politics and ethics to how easy generative tools are to use, how it makes them feel, and how they see themselves in the future.

During a period of ten days, a total of 42 Liipers answered the survey, with responses across seven role types (see fig.4). The focus of my data analysis was the topic specialists, the title given to designers.

ROLE TYPE	Nº
Agile Specialist	1
Strategist	1
Business Developer	3
Business Support	4
Topic Specialist	5
Product Owner	7
Software Developer	22

Figure 4 -Table of survey answers categorised per role type.

3.1.4 Thematic Analysis

This study follows the six-step thematic analysis framework of Braun & Clarke (2006) to analyse the data. This method aims to identify patterns in the data to then describe them in detail. To do so, different codes are created to define themes based on the relevance, occurrence, and size of relevant insights in the data set. Between semantic and latent levels, I have taken a semantic and realist approach to data analysis, defining the explicit ideas expressed throughout the interviews, observation, and survey. In other words, I have compiled what was transmitted by the employees with little to no analysis of underlying or implied assumptions.

Divided into six phases, this framework includes the following steps: “Familiarising yourself with your data”, “Generating initial codes”, “Searching for themes”, “Reviewing themes”, “Defining and naming themes”, and “Producing the report” (Braun & Clarke, 2006, p. 87). It starts

by getting immersed in the data; re-reading and annotating important ideas followed by highlighting initial points of interest and collecting data that fits the different codes created; agglomerating codes into themes; reviewing if the themes work together with the codes and the general data set; continuous iteration; refining and defining the names of the themes, and finalising with the creation of the report (Braun & Clarke, 2006). In my process, steps three, four, and five underwent multiple rounds of iteration, hence not following a strict linear timeline.

Stage 1 - Familiarising yourself with your data

The analysis started by reading and highlighting all notes taken during observation, conversations, AI talks, interviews, and from the survey's responses. Due to the lack of full transcriptions of the conversations, the information used was based on the extensive notes I took during those and my own memory of the interactions, as I repeatedly consulted and read my notes while choosing and reading the literature. It is also important to note that after the survey was conducted, an initial issue map was created, and although it was eventually replaced, the analysis remained relevant as it also aimed to separate the main issues of Liipers into themes and codes (see appendix A).

All the information was pasted into a Figma file where the data could be compared, which resulted in a considerable number of blocks of information that were subsequently analysed, ignoring irrelevant ones and synthesising and coding the rest (see figure A in appendix B).

Stage 2 - Generating initial codes

The codes were obtained through a majority semantic approach to define the superficial ideas or the source data with a realist epistemology. That is, the codes represent straightforward feelings, motivations, and experiences present in the source data.

Coding all the information took multiple iterations to make sure the resulting codes were coherent in language and format, as well as avoiding duplicates by merging or eliminating unnecessary nodes. For that, I re-read the data set and the information blocks to make sure that the meaning of each code was not lost and that the main ideas were still present. To maintain coherence, unrelated comments that were only mentioned once by a single person were not considered relevant for the overall analysis. At the end, I was left with around 90 "digital post its".

Take figure 5 as an example of the coding of a data extract from the survey and figure 6 as the coding example of a data extract from conversation notes. Each extract leads to one or multiple codes.

Data extract	Coded for
"All studies show that the more we use AI in general, the more we lose our core and hard learned skills. It is debilitating. So far it is only destroying jobs, the economy, and the planet."	risk of excessive dependency on AI tools
	fear of AI-driven job displacement
	perceived negative impact on economy
	concerns about environmental footprint of AI use

Figure 5 - Example of data excerpt from the survey and codes used

Data extract	Coded for
"Sometimes AI makes the process harder when it has a lot of bugs and mistakes. Especially when it does not understand what the user wants it to do to"	AI errors increasing workload
	high correction cost when AI produces wrong outputs
	AI requiring time to comprehend prompts

Figure 6 - Example of data excerpt from a conversation and codes used

Stage 3 - Searching for themes

During stage 2, some subconscious groupings started to appear. For this stage, the previous organisation was scrapped to give way to a blank page to analyse the codes further and discover underlying themes. By doing this, I discovered a high number of duplicate codes, for example "give low-value tasks to AI" and "delegation of mundane tasks to AI", which meant the same thing. It was clear that another round of analysis of the codes was needed to weed out any redundancies and facilitate the development of clearer themes that did not overlap.

The result of this first analysis was six themes: Needs for AI adoption, AI capacity limitations, AI benefits, Professional fears and anxieties, Ethical concerns and AI impact in the market; with 13 sub-themes in total (see figure 7), see figure B in Appendix B for the codes in each group. Some nodes were transformed into sub-themes, such as "fear of job replacement" and "time efficiency".

It is important to consider that this thematic map is the result of previous iterations where I encountered multiple issues: the themes' titles were descriptive and not abstract enough, the themes were too vague and there were not enough sub-themes.

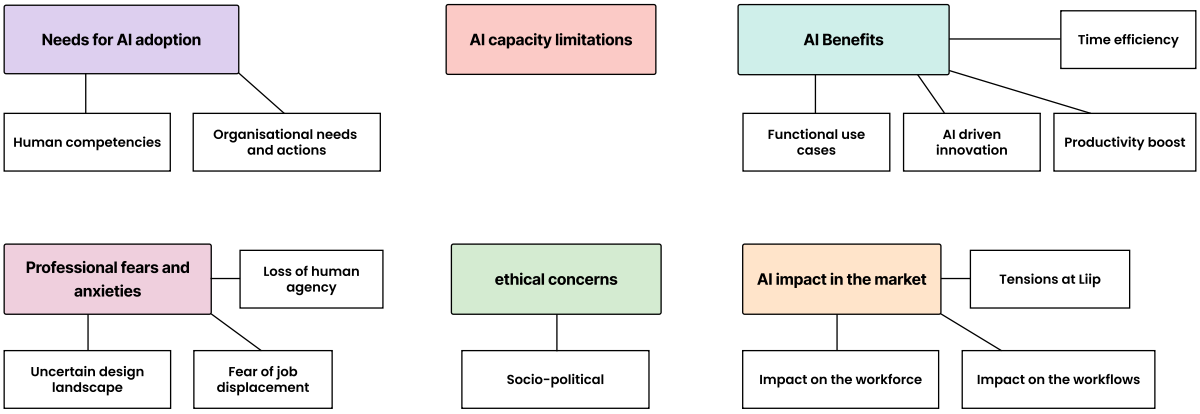


Figure 7 - First thematic map

Stage 4 - Reviewing themes

After carefully reviewing the themes and subthemes, there were connections that were either too vague, unnecessary, or too constricting. It was also necessary to rephrase the title of themes and subthemes, so they met the meaning of all their nodes and were more explicit.

Stage 5 - Defining and naming themes

Finally, the resulting thematic map represents seven themes with ten subthemes total (see figure 8), see figure C in Appendix B for the codes in each group. Due to the complex nature of AI adoption processes and the psychological impact it has on Liipers, the subthemes are important for clarity and the merging of one or more blocks would result in loss of context and necessary insights. Some necessary changes included renaming “AI impact in the market” to “Organisational and market pressure” and “Professional fears and anxieties” to “Professional identity under threat” as well as making “Losing agency to AI” as its own theme and merging “Socio-political” with the rest of ethical risks.

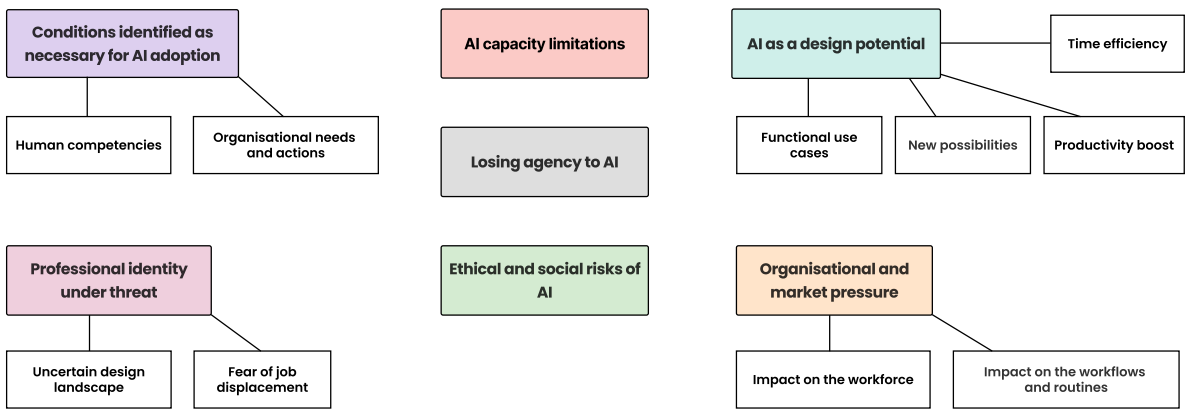


Figure 8 - Final thematic map

This map showcases the primary ideas present in the data set, which depict Liiper's behaviour and mentality towards AI and allow for an overview of the current situation.

Stage 6 - Producing the report

Between all the themes, a central friction emerges: While designers recognise AI's potential, they feel personally, professionally, and organisationally unprepared to handle it.

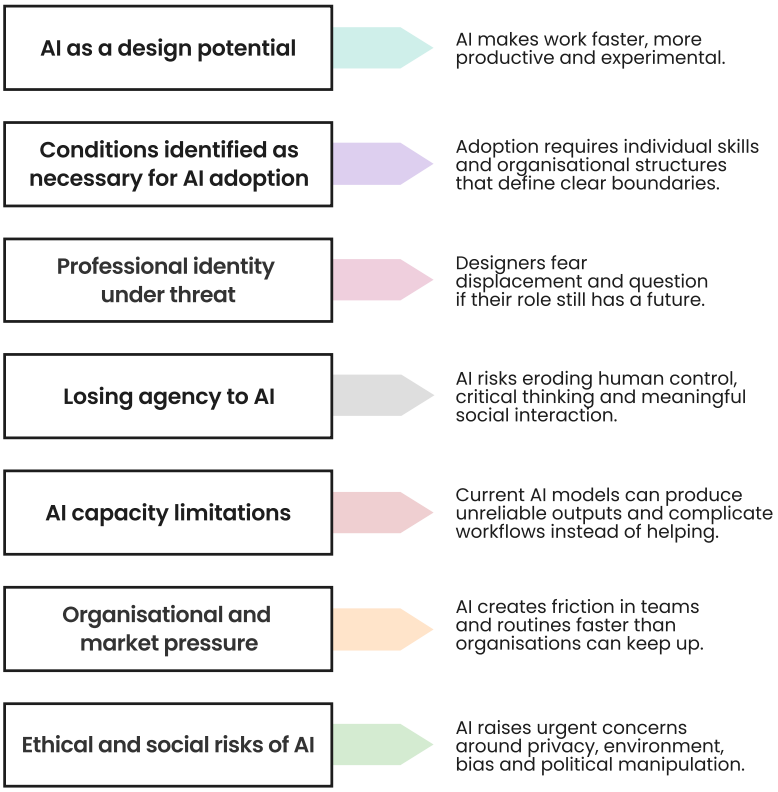


Figure 9 - Themes and theme descriptions

In this stage, each theme will be described and supported by excerpts of the data set and literature from the theory chapter. The experts used include quotes from the survey response pool and ideas shared during conversations that have been paraphrased.

AI as a design potential

This theme contains all the opportunities and benefits Liipers feel like machine learning brings to their jobs and design. It allows creators to work faster and save time to work on more important tasks while AI handles mundane, repetitive tasks. It also highlights how this technology expands the current scope of design tools and recurrent use cases (make reports, analyse data, brainstorm, explore ideas, etc.) and opens new possibilities to creatively experiment and change the way we work.

"I will be able to complete tasks that used to take me several days in just half an hour."

It is understandable why one of the biggest selling points of deep learning technology is how it significantly cuts down the time needed to complete certain tasks. People are constantly looking to optimise their processes to both fit bigger and more interesting projects into their repertoire and try to gather some resting time for other activities outside the workplace or mandatory tasks (Chanchamnan et al., Yu, 2025).

"Save time on debilitating tasks, so that I can use the remains in a better way."

Machine learning gives people the ability to significantly cut down working hours necessary for each task, mainly repetitive, low-stakes ones that take too long to complete. By allowing a new allocation of tasks where the machines do the "boring tasks", designers can focus their skills and cognitive energy on doing things they actually enjoy (Yu, 2025).

"Read documents for me, search for insights in transcripts of interviews to help me save some time processing interviews. (...) I can also build a prototype with it"

AI offers several use cases for day-to-day life that are being implemented by designers. Research, ideation, brainstorming, and prototyping are just some of the tasks that machine learning can augment and increase productivity, save creatives time, and introduce new, unique ideas (Chanchamnan et al., 2022; Lee et al., 2025).

"AI allows us to reinvent the tools, the workflows, and how we work together"

Beyond current use cases, the technology opened new possible pathways by generating new tools, introducing innovative platforms, and redefining what work means and how machines and humans collaborate. Although it is still in an experimental phase, the future of AI-powered design could look very different with the integration of tailored toolkits for creatives.

Conditions identified as necessary for AI adoption

Adoption does not exist in a vacuum. Without the necessary skills and knowledge of the user and considerable efforts made by businesses to facilitate the use of AI and mitigate misuse, it is not possible to aim for a responsible adoption of AI. This theme identifies these two pillars as required conditions for the integration process.

"I find it very pleasant not to have to figure everything out and research everything myself, but to be able to benefit from the experiences of my colleagues."

Collaborative learning through peer-to-peer formats has proved to be a successful way to spread know-how and knowledge throughout Liip. As resistance is a common defence response to what people do not understand and is forced upon them, having designers learn from each other created a welcoming environment for experimentation and failure without consequence or judgement. Using co-design, designers can create their own initiatives and workflows (Steen et al., 2011).

"People should not use AI without understanding it first"

One of the most relevant requirements is a basic knowledge of how LLMs and GenAIs work, what and how to prompt correctly, be prepared to be critical of generated outcomes to mitigate hallucinations, etc (Santana & Díaz-Fernández, 2023). Starting out with no basis just ends up with disappointment, frustration, and low-quality results.

"Start with smaller things, easy, low-risk tasks to use AI in"

Experimentation is a given when talking about adopting a new technology. It is necessary that designers have safe spaces to try out new things, test the limits of the tools, and take risks. Allowing designers to take creative liberties in smaller tasks like landing pages or even contact forms is a good way to slowly introduce AI as a co-author in bigger projects (Al-Kfairy, 2025).

Professional identity under threat

As machine learning systems seep into the market and influence how people work and what they work on, a pushback is felt from the workforce. Being introduced to a technology that is sold as an everything tool that can analyse and create anything can and is being understood as a replacement for designers and other positions. This leaves workers to figure out new alternatives, broaden their skills, or even change jobs to keep being competitive and maintain the resources they need to live.

"In less than 3 years, it already killed the momentum in my branch (design)"

The quick evolution of deep learning networks and the growing hype around the technology have led some people to conclude that design and the arts are "dead". This misplaced assumption comes from the idea that because GenAI programs can generate images and brands in a

few minutes, designers are not needed anymore and will be replaced (Dwivedi et al., 2021). This belief, although based on wrong ideas, has unfortunately steered some people away from creative disciplines and has left designers in an awkward position (Garcia, 2025).

"AI is redefining our jobs; we need to redefine the future"

Designers are beginning to feel the need to redesign their own futures by continuing to broaden their skills in design and coding to avoid being replaced or becoming obsolete (Garcia, 2025). AI is allowing people to operate outside their own roles, with many designers being asked to code and developers using AI to formulate designs. This uncontrolled overlapping of roles can create friction between teams as they overstep into each other's domains.

"There are no more opening jobs for junior designers, the school in which I taught UX design just close this course"

Although this is not happening everywhere and there are still a lot of art courses and educational institutions, seeing some of them disappear impacts creatives' confidence in their skills, their roles, and their futures. It is increasingly harder to find junior positions as the smaller and easier assignments they would be tasked with are being given to AI agents. Additionally, with how quickly the technology evolves, what designers are learning during their education is outdated by the time they graduate (Bowles & O'Hanlon, 2025).

Losing agency to AI

As we delegate more responsibilities to machines, start thinking through machines, make them read and make things for us, we feel like we are losing the upper hand against the technology. Especially when considering generated personas and automated user research, we see human social interaction be undermined by machines and isolate people with their computers.

"People need to accept we will lose control using AI"

When using artificial intelligence as a co-author, it is hard to delineate boundaries. The agents can be hard to guide and end up doing something that does not correspond to the idea the designer had in mind (Zhang et al., 2023). In contrast, if the designer wants to have all the decision-making power, the AI is not very beneficial and feels just like an extra step. The idea that to properly gain from the use of these tools, designers need to give away their creative control and accept less than ideal results from the machines (Yu, 2025) is incredibly concerning

for some creatives who are afraid of losing their freedom and compromising the quality of their work.

"AI tools can often take away from people's autonomy and creativity"

As mentioned in the previous excerpt, using generative tools and other programmes to produce design outputs requires a level of compromise of control that people are not yet ready for. It directly impacts their autonomy to make their ideas come to life however they want to, and it can even hinder their creativity when they must adapt their workflow around the tools and accept the limitations that come with using this technology (Garcia, 2025).

AI capacity limitations

There are two sides to every innovation. It is true that AI has opened a multitude of possibilities across industries and facilitated a lot of tasks, but it is also true that the technology is not perfect and there are a lot of downsides to its use. From incoherent outputs to agents that lack autonomous decision-making and the responsibility of having to double-check every single result for hallucinations, sometimes machine learning is more inconvenient than helpful.

"Acceleration trap – sometimes the more mundane work keeps you sane"

Although it is possible to achieve productivity gains by automating certain assignments, the goal of making things increasingly faster is not sustainable. The paradox of AI speed characterises how when everything is automated, workers will be expected to work more rather than gaining free time. The fact is that true productivity is not achieved by making everything faster but by organising teams better, respecting workers' mental health to avoid burnout, and creating a stable strategy (Bowles & O'Hanlon 2025). People need simple, low-effort repetitive tasks to maintain a sense of accomplishment, not overload their cognitive capabilities and save their energy for high-effort projects.

"Sometimes AI makes the process harder when it has a lot of bugs and mistakes. Especially when it does not understand what the user wants it to do to"

A major concern of deep learning networks is hallucinations, lies, and misguided information that is presented as factual (Ahmad et al., 2023). Often as a result of a compromised data set or simply due to the nature of the technology, it can result in bugs, mistakes, and weird outputs that require the users' time and effort to try to fix. In complex tasks where these mistakes are more likely to happen, AI becomes an obstacle instead of a facilitator, especially when users

do not understand how to prompt with enough context and the machine does not understand the goal of said prompt.

Organisational and market pressure

Beyond technical limitations, AI impacts the work environment, the workers, the companies, and the general market. The adoption of the technology feels like a race with everyone trying to gain a competitive advantage. This results in an overarching feeling of stress and fear of falling behind, companies struggling to find the best way to implement new systems and reorganise their teams, and an experimental phase where it seems like AI is adding a layer of complexity to every process.

"[where the pressure is coming from] Liip & everything that we read online pushes us to believe that only the ones mastering AI will survive"

Designers are feeling the pressure to adopt AI in their workflows, whether they prefer it or not. This pressure is not always direct but felt through everyday interactions and media where seemingly everyone is talking about how they use the technology and how it is the future of the industry. A lot of times, this push is sold as a way to gain an advantage over other professionals, and it instils an idea that everyone is suddenly competing to maintain their jobs and their careers. In a company, the pressure is elevated as people are expected to accept the new systems' change and help represent the firm as an innovative participant in the market (Bughin, 2025).

"The role of the designer is moving into something else: a review and a client. AI is one more middleman in the project."

With generative tools being able to produce high-quality renders, images, videos, and songs, helping produce prototypes and brands; and LLMs having the capacity to write stories, scripts, and marketing campaigns, the role of the designer is changing. Moving to the passage seat, it will be possible for designers to become the arbiters of the creation and not the main creator (Garcia, 2025; Yu, 2025). While it is important to note that it can only happen when the situation calls for it and that it is not a mandatory practice, even with smaller tasks, designers are already acting ad overseer of AI-generated outputs. In a sense, this means they are also acting as the clients themselves, consuming a product made by artificial intelligence. It is understandable how designers feel the need to resist this change, as it essentially automates the favourite part of their work, creating, and it poses the question of how far the automation can go and if it will ever reach the level of human creativity.

With the Holacracy model we need to contribute to our personal skills, the internal company, build teams, (...), with limited time. It is not possible"

Holacracy is the governance model used by Liip, where management and leadership are decentralised and employees are encouraged to be autonomous, create their own initiatives and propose their ideas to the company. While this model allows for a fairer workplace, highlights transparency, focuses on purpose and is highly agile, qualities that are beneficial for a successful adoption of AI, it can also have a negative impact on the workers. Having the opportunity to contribute not only to their personal growth and side projects but also to the company and its internal teams are responsibilities that can weigh on designers' shoulders and act as a hindrance to their disposition to add more responsibilities to their schedule. With little time to explore tools or find external collaborators, designers might feel overwhelmed by the openness of this governance model and would benefit from some stricter limits in certain aspects and contexts.

Ethical and social risks of AI

Outside of the office and at a global scale, AI has an impact on society, politics, the economy and the planet. A complex topic, the ethical concerns of the use of the technology have only increased as more and more issues start to arise. Biases in the training data are repeated and exaggerated through chatbots and LLMs, data centres use millions of litres of water to cool down their servers, political misinformation and fake news are created to influence elections, fear-monger and scam people and those who fully benefit from the proliferation of the technology are the CEOs of AI tech companies and millionaires who do not always have the population's best interest at heart.

"Perpetuating gender bias"

Even with a careful curation of data sources, there are biases we cannot predict and are easy to miss. With that, in a normal situation, it is more common that the data sets have not been thoroughly examined, and the biases present are spread through language models in sexist, racist, and homophobic comments, among other discriminations (Dwivedi, 2021; Garcia, 2025). Often taken at face value, a lot of the outputs are not fact-checked or critically analysed and are shared between friends, social media, and even the news, which causes a lot of people to avoid using them. In design, biases are a huge risk in projects and can compromise research and user testing, or the hiring process of future designers (Bowles & O'Hanlon, 2025; Down, 2025).

"(technologies) are controlled by technological oligarchs who only have power and money at heart, currently fuelling a fascist overcome of the USA."

The current majority of most used AI platforms and investors belong to companies from the USA. Businesses like Microsoft, Alphabet, Amazon, and Meta, owned by immensely powerful and rich men, are sometimes affiliated with certain political movements that people would prefer not to support (Dwivedi, 2021). At Liip, a Swiss company, the political aspect is not lost on people, several of whom are concerned about what their use of the technology is fuelling and distrust the intentions of the institutions who establish them.

Overall analysis

The seven themes come together to tell a coherent story: Liip, as most other companies, encountered AI before having all the conditions for a smooth transition. "AI as a design potential" shows that the value of the technology is being recognised and that resistance does not come without reason. However, the rest of the themes express everything that is preventing them from embracing it, from technical limitations in "AI capacity limitations" to bigger social implications in "Ethical and social risks of AI". In "Conditions identified as necessary for AI adoption" we understand that neither individual competency nor necessary organisational structures are developed enough to support a full-scale adoption strategy. We also observe, in "Organisational and market pressure" how the transformation is being fuelled by competitive pressure and how the current model has revealed tensions in the companies' processes. Both "Professional identity under threat" and "Losing agency to AI" expose the human cost of these tensions, where designers feel like they are being asked to delegate control to machines and fear becoming obsolete if they refuse.

At the end, it is possible to see a company in its experimental phase, having awareness of what needs to be done to get closer to its goals, but struggling to realise their vision without setbacks and hostility. The main barriers are not technical but structural and human. Perceived lack of adequate guidelines, erosion of identity and agency, contradicting morals, the pressure for speed and a governance model that needs adapting overshadow hallucinations or the high correction cost of generated outputs. For Liip, the way that its Holacracy model shapes transformation, both as a limit and as an opportunity is a major and actionable question for the organisation to address, since it is an internal factor specific to the firm. Regarding the rest, it is clear that before designers can welcome AI with open arms, a list of conditions needs to be implemented first to promote a healthy, productive environment: ethical and structural clarity, job

security and psychological safety, and institutional guidelines for education, governance, AI use and problem-solving.

3.1.5 Results

Through the research and subsequent findings, an outline of the current situation of Liip can be defined. Liip is the exploratory phase of AI adoption, characterised by uncertainty, small pilot projects emerging within teams, a trial-and-error approach to defining new values, guidelines, and toolkits, and a mix of apprehension and excitement across the board. While the agility of its governance model and the collaborative mindset present a strong base for the transformational change, the research reveals that designers are still reticent to welcome the change fully. This situation is fuelled by the perceived lack of guidelines, competitive pressure, and the psychological impact the technology has had.

Finding 1

AI is intentionally being adopted as a bottom-up dynamic, without strong top-down / company-wide orientation. With individual experimentation and solutions emerging from small circles, people feel both excitement towards the innovation and the freedom to try it, and anxiety from being in an unknown territory with not enough support.

Finding 2

The conditions for a responsible adoption are partially present but not formalised. Small initiatives and collaborative learning groups exist but are not yet connected or well known to every Liiper, causing knowledge gaps between them while the processes are ongoing.

Finding 3

The technical limitations of AI matter less than the psychological toll it causes. Concerns over loss of agency, role replacement, and the future of the design industry act as the primary barriers for a smooth adoption and should be the first to be addressed.

Finding 4

While Liip has taken initial steps toward ethical AI use, including data collection clauses in client projects and a preference for open-source tools, the ethical concerns expressed by par-

ticipants expand beyond these factors. Environmental impact, political affiliation, the spreading of biases, and the general morals of using a technology that can invade people's privacy and data suggest that future frameworks are still needed to handle these issues.

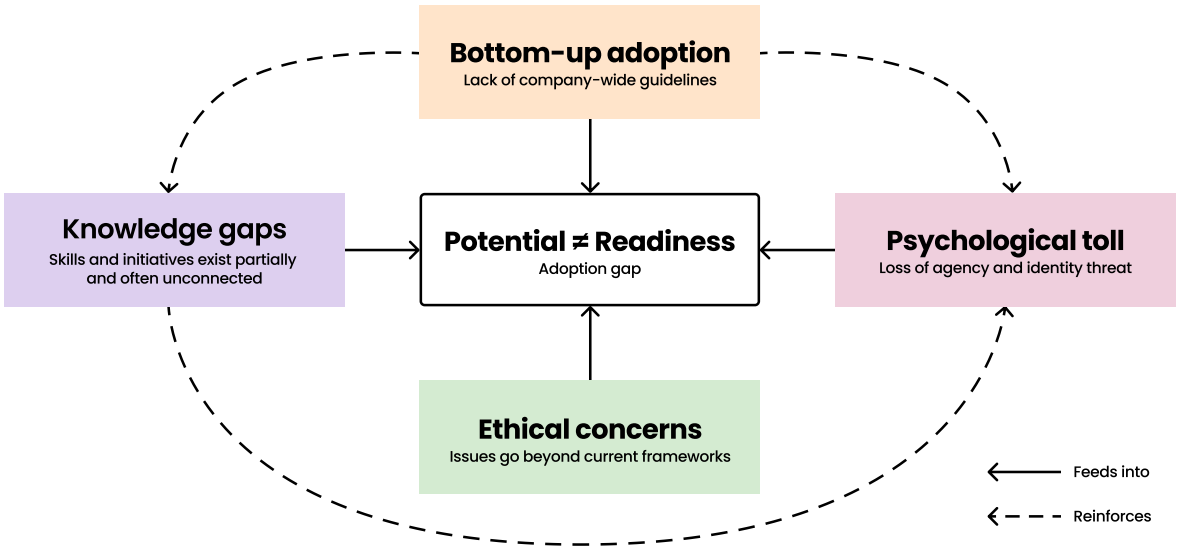


Figure 10 - Issue map based on findings

Together, these findings showcase a company caught in mid-transition, enthusiastic about the potential of AI but structurally unprepared. The missing link is not awareness but defining boundaries and rules that give way to a confident and coherent practice.

3.2 Prescriptive Study

The goal of the PS section is to answer the sub-question "How can processes be adapted or redesigned to support human-AI co-creation in organisational design contexts?". It is in this section that the ideal situation is defined based on the findings from the previous phase. It describes the design of the process of exploration, ideation, and iteration until the final resulting collection of strategic initiatives.

3.2.1 Exploration

Attempting to answer the main issues previously identified in Figure 10, this exploration will define the optimal scenario for AI adoption at Liip.

In an ideal situation, Liip would transition from its current experimental phase to a grounded model state of AI integration supported by clear but agile structures that continue to value collaboration, transparency, and autonomy. Explicit guidelines would be in place to reduce

ambiguity during the design process and client projects, and designers would have access to tools that fit their specific workflow needs rather than generic solutions and off-the-shelf products built for the general public. With dedicated R&D time, protected and implemented into the schedule, designers would be able to experiment at a small scale, co-create their own guidelines for use, and find a way to collaborate with the technology without losing the essence of their work and identity: creativity, imagination, empathy, curiosity, and contextual awareness. Knowledge would move faster across the company, with small team workshops happening concurrently with larger ones so that knowledge found in tight circles quickly reaches everyone. Ethical concerns would take a more central role, where solutions would be created through an ongoing collaborative effort, one that evolves as the technology changes its doings. At a larger scale, external collaborations, established toolkits, e-learning tools, and continuous training opportunities would be available across teams as a foundation for professional growth that directly targets the fear of being left behind.

With the goal of integrating "AI as a co-author" in Liip's design teams in a way that enhances human creativity rather than replacing it, there are multiple objectives that must be worked on simultaneously:

Objective 1 - Establish guidelines for human-AI collaboration

Taking away decision-making pressure from individuals, clear boundaries that define how AI enters the design process are set by the company and lay a foundation that everyone can rely on. This objective does not aim to limit experimentation but structure what tasks AI will take over, which ones will be undeniably human-made and how the collaboration is communicated transparently between teams and with clients.

Objective 2 - Build collective AI literacy

Knowledge at Liip concentrated itself in groups and still takes some time to spread company-wide. This objective aims to connect what already exists and formalise projects that are already working, such as small circle workshops and individual employee initiatives. The goal is that every designer understands the possibilities while also being critical, to be able to engage with AI with confidence. Instead of an independent trial-and-error approach, a collaborative strategy offers mentorships, tailored training, toolkits and workshops.

Objective 3 - Promote professional growth and designer identity

To address the biggest barrier of AI adoption, this objective targets designers' biggest concerns and reframes the designer-AI collaboration. With a focus on human capacities such as empathy, creativity and contextual awareness, this goal aims to highlight the value of the designer by establishing Human Capability standards, elevating their skills and motivating them through certifications and creative challenges. Like this, creatives will be able to keep up with the evolution of the technology and confidently defend their identity as designers.

Objective 4 - Practice ethically aware AI use

While Liip has already worked towards a responsible use of AI, with data use guidelines in client projects and a preference for open-source tools, there are still things to be done. Addressing environmental impact, political affiliations and bias, this objective aims to establish an ethics board or system and risk assessment processes to ensure ethical awareness is considered in practice and not just as an afterthought.

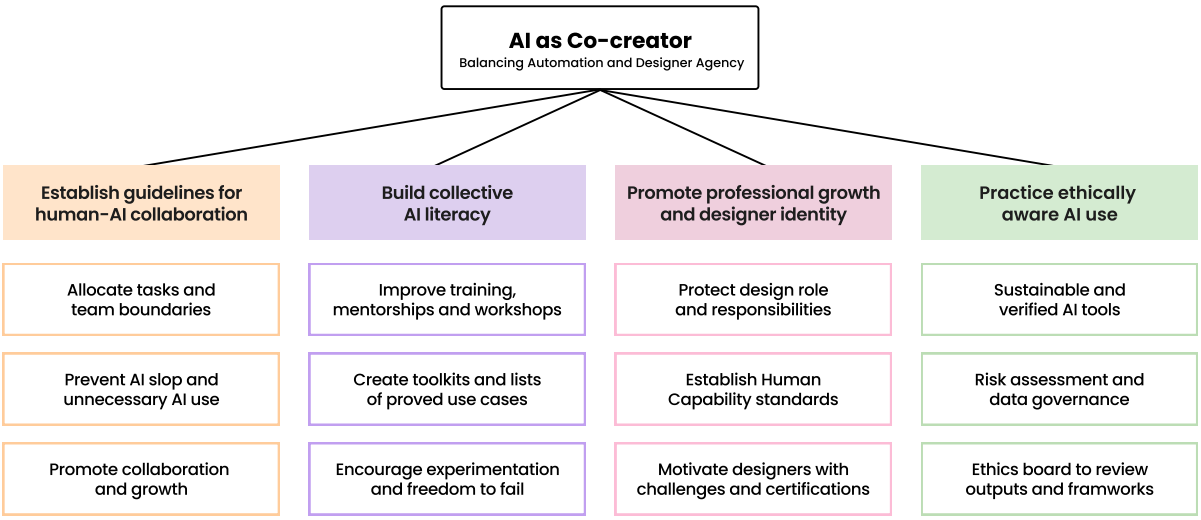


Figure 11- Objectives tree

3.2.2 Ideation

The ideation process explores the creation of adoption strategies based on the DS1 and PS phases. A Figma file was the primary method of compiling, organising, and designing the strategies as well as supporting images and diagrams used throughout this study (see Appendix D).

The initial ideation (ID1) followed an earlier version of the objectives tree (see Appendix C) that was created based on the earlier version of the issue map (see Appendix A). After the thematic analysis, it was clear that these maps did not accurately represent Liip's challenges and needs and were updated through a careful consideration of the data and feedback from designers. The current objectives (see figure 11) directly address the DS1 findings and ground the ideation in empirical evidence. Keeping the previous ideas in mind, the following ideation processes document how the new defined objectives were translated into strategic initiatives.

As a starting point, all the literature in Chapter 2 was carefully revised to identify strategies and relevant ideas present. If relevant to Liip, these were introduced into the Figma file before being adapted to Liip's context and organised into different groups. Additionally, new ideas were created based on the literature and the empirical research and grouped accordingly. At the same time, three different ways of grouping the initiatives were chosen to convey the ideas clearly and be useful to Liip: according to the objectives tree; divided into organisational, design team, and individual levels; and following a timeline roadmap.

Under each objective, "how can" questions were added to support the creation of new ideas and the assignment of existing ones to each goal:

Establish guidelines for human-AI collaboration

- How can AI and human tasks be separated and defined?
- How can Liip make AI use consistent across teams and client projects?

Build collective AI literacy

- How can knowledge present at Liip reach every designer?
- How can Liip access and integrate external AI knowledge into its own practice?

Promote professional growth and designer identity

- How can designers go from reticent to AI experts?
- How can Liip protect the designer role?

Practice ethically aware AI use

- How can ethical criteria be included directly in design decisions?

- How can Liip prevent bias and misinformation?

Following that, ideas started to be assigned per goal, while removing duplicate or similar insights and initiatives that do not fit Liip, either because of its governance model (e.g. “create and ethics advisory board”) or because it has already been implemented (e.g. “define data use policies in projects”). This resulted in initiatives such as: “Create a NO AI SLOP guideline to prevent uncritical use of AI” under *Establish guidelines for human-AI collaboration; “*Continue Liip talks where designers share what worked and what did” under *Build collective AI literacy*; “Motivate designers with AI skills certifications” under *Promote professional growth and designer identity* and “Create risk assessment analysis processes” under *Practice ethically aware AI use.*

3.2.3 Iteration

Before reorganising these ideas, they were reconsidered and evaluated. Some loose ideas from the ID1 were revised and merged with existing ones, for example, “Strive for collaboration and growth instead of just efficiency” + “Communicate the narrative with Liipers, collaboration and elevate human value (ID1)” = “Communicate a narrative that prioritises collaboration and growth instead of just efficiency”, under *Establish guidelines for human-AI collaboration*.

Other changes including rephrasing some ideas such as changing “Start projects by allocating AI and human tasks” to “Start projects determining AI-eligible tasks”; assigning ideas to different objectives such as moving “Use AI as a powerful tool to combat design fixation” and “Assign new tools like sketch to image or 3D generators to different designers to explore each quarter” from *Build collective AI literacy* to *Promote professional growth and designer identity;* and adding new initiatives that either complement others or fill an opportunity gap between different initiatives such as “Promote a critical reflection exercise after projects to document the creative limitations of AI” under *Promote professional growth and designer identity.* This initiative added onto others such as “Continue Liip talks where designers share what worked and what didn't” under Build collective AI literacy with a focus on the creatives’ critical view over AI and on highlighting the need for designers.

3.2.4 Structuring

After the refining of the strategic measures in each objective, they were assigned to different levels: organisational, design teams and individual designers, through a straightforward process. Objectives including “Establish guidelines for human-AI collaboration” and “Practice ethically aware AI use” naturally provided most initiatives in the organisational level, while the other two were divided between design teams and individuals. Subsequently, they were ordered according to an 18-month timeline that delineated a three-phase AI adoption roadmap.

When building the structure with levels, certain ideas had to be adapted for the bottom level “Individual designers”. For instance, “Incentivise Liipers to be vocal about ethical issues” is present in design teams but in individual designers it reads like “Be vocal about ethical issues”. With this, some initiatives were applied to more than one level. The goal is to define who is responsible to make each idea work and show how the top levels influence the bottom ones.

For the roadmap, a timeline of around one year was chosen due to the evolving nature of artificial intelligence, although the last phase goes on indefinitely. Divided into three blocks, the first one occupies the first three months and characterises the “Build the foundation” stage. It defines the norms and principles that need to be established before anything else, do not require resources or big changes and mostly concern mindset and behaviours. Next, the “Implement AI in practice” stage takes the following six months with measures that will imply active coordination and some level of change. It includes infrastructure needs, tools, training, processes and other documents that will be necessary to support the behaviours of phase 1. Lastly, phase 3 aims to solidify the benefits from foundation, scaling successful strategies and embedding experimental pilot projects. “Scale what works” requires systematic transformation, external partnerships and big structural decisions, hence why it takes a minimum of one year but with no time limit. The aim of the roadmap is to clarify which strategies need to come first to create a strong base and delineate how the adoption should evolve.

3.2.5 Strategies

Liip already fills out a lot of the requirements needed to begin a responsible adoption of AI. Its Holacracy model showcases their innovative mentality and allows for Liipers to make decisions together without strict management preventing them from trying new things, and the AI changemakers’ plans are pushing the company on the right track. The goal is not to force AI adoption but to encourage people's own curiosity and guide them through a big change. There

are also some guidelines in place, such as suggested tools to use and data concerns over quality and accuracy. The Ecocycle model is used by the changemaker to visualise the relevance of the current system in its various components and the need to rework them individually or not, be it the infrastructure, competencies, processes, etc. Other models (such as ADKAR) are used to facilitate the change as well.

With this in mind, the following strategies aim to continue the efforts being done at Liip and suggest a way forward that respects Liips' values, Liipers' wants, and what is feasible for the firm to achieve.

3.2.5.1 Assigned per objective

The organisation of the initiatives to address each objective defined in the PS phase seeks to respond to the designers' main needs (see figure 12). Objective 1 contains strategies that highlight the limits of AI use and make sure everyone is on the same page. New values are communicated and enforced by guidelines that respect designers' creativity and facilitate collaboration between teams and humans and machines, while efficiency and productivity are still a focus but maintain a secondary role. By delimiting where AI use can go and which tasks are ready to be automated, designers will not feel the creeping fear of being immediately replaced and will feel more comfortable experimenting.

Objective 2 builds on that comfort and offers a more supportive environment for exploration without fear of failure. With a focus on training tools and programmes, the strategies also open the possibility of external collaborations to access both knowledge (e.g. see how young designers are using AI in innovative ways and how universities are addressing AI use in creative disciplines) and infrastructure (e.g. new technology, discounts in maintenance, access to beta tools, etc.). Most importantly, beginner-friendly pathways that take beginners to the same AI literacy level as advanced users through smaller and more accessible assignments.

Developing designers' skills is even more prominent in Objective 3. Here, the creative workflows are the main focus, giving designers designated time for individual experimentation, ideas on how to use tools to their advantage, exercises to increase motivation, and suggesting behaviours that elevate existing ones at Liip, such as fostering peer recognition and starting projects from uniquely human ideas.

Finally, Objective 4 touches on ethics, legal policies, risk management, transparency with employees and clients, and mitigation of bias, hallucinations, and other issues brought by AI. Although harder than the other objectives, as it depends on the development and ethical choices of external products, these ideas serve as guides for the future, and the organisation should do what it can with the tools available now.

AI as Co-creator

Balancing Automation and Designer Agency

Establish guidelines for human-AI collaboration	Build collective AI literacy	Promote professional growth and designer identity	Practice ethically aware AI use
How can AI and human tasks be separated and defined?	**How can knowledge present at Liip reach every designer?**	**How can designers go from reticent to AI experts?**	**How can ethical criteria be included directly in design decisions?**
How can Liip make AI use consistent across teams and client projects?	**How can Liip access and integrate external AI knowledge into its own practice?**	**How can Liip protect the designer role?**	**How can Liip prevent bias and misinformation?**
Use the double diamond method as a guide to assign AI automated tasks, augmented tasks and NO AI tasks	Create an internal platform with curated tools and prompts tailored for design projects and tasks	Create creative AI challenges and display the results from the designers. Not as a competition but to showcase what can be done	Develop a simple ethical checklist for designers to consider before using AI in client projects
Define new Liip AI values, what the company stands for and against. Highlight what will not be replaced	Make ready to use toolkits and guides about AI use, do's and dont's, use cases, etc.	Highlight the need for creativity, ingenuity and imagination in AI-augmented design projects	Incorporate Corporate Social Responsibility (CRS) models intro current model at Liip
Explore AI generated guides that define processes, define limits and adapt to each project	Let designers explore automation starting from mundane tasks	Prioritise human interaction during research and user testing, even if augmented with AI personas	Embed hallucination control processes in every instance of AI use
Communicate a narrative that prioritises collaboration and growth instead of just efficiency	Collaborate with universities and research institutions to get new perspectives and insights	Open pathways for designers to contribute to AI tool development	Establish a rotating multidisciplinary AI revisory board
Define efficiency-boosting tasks to be automated	Offer tailored training and e-learning tools	Avoid pushing designers to an "arbiter" role with no need	Promote critical thinking and analysis of AI outputs
Create a NO AI SLOP guideline to prevent uncritical use of AI	Use a collaborative, participatory approach to AI learning	Challenge creatives to use AI in unique ways and think bigger	Work closely with the legal team to create legal frameworks
Start projects by allocating AI and human tasks	Foster inter-team workshops to build collective knowledge	Motivate designers with AI skills certifications	Use tools with less ecological footprint

Create guidelines for designers and developers to align their workflows and prevent unwanted overstepping	Create catalogue of previous and ongoing individual AI explorations to document what has been/can done, including prompts	Use AI to create a designer availability schedule to optimise project teams and protect designers time	Implement bias awareness training and processes of review
Apply human-in-the-loop framework and other practical AI-human collaboration methods	Continue Liip talks where designers share what worked and what didn't	Assign new tools like sketch to image or 3D generators to different designers to explore each quarter	Use tools from less ethically ambiguous sources
Develop a shared glossary of AI-human collaboration terms to facilitate communication between designers and developers	Develop beginner-friendly pathways to address knowledge gaps between them and advanced AI users	Foster peer recognition for human-led creative decisions and contributions specifically	Incentivise Liipers to be vocal about ethical issues
	Explore AI generated real-time coaching tailored for designers' needs	Adopt the Human Capability Standards (HCS) framework	Create risk assessment analysis processes
	Collaborate with stakeholders to access technology and knowledge	Establish R&D time for exploration protected in the schedule	Focus on ethical decision-making and not just on outputs
	Prioritise prevention and handling of AI hallucinations	Do not define how designers should work but offer support	Promote transparency about AI use in projects
	Pair less confident designers with more enthusiastic ones	Always start projects with human ideas that can be expanded with AI	
		Use AI as a powerful tool to combat design fixation	
		Promote the value of human made work and ideas	
		Customise HRM practices in accordance with employees' competencies	
		Prioritise skills and competencies over role titles	
		Promote a critical reflection exercise after projects to document the creative limitations of AI	

Figure 12 - Strategies assigned per objective

3.2.5.2 Assigned per layer

The same strategies in Figure 12 are now reorganised per layer, going from the organisation at the top to the designers at the bottom. The idea is that things need to change on an institutional level to open space for smaller strategies from the design team and individuals. Taking a look at Figure 13, it is possible to see how the number of initiatives diminishes along the levels, demonstrating how critically important systemic change is for the success of AI adoption in specific teams.

This top-down layout of ideas also serves to address Finding 1 from the DS1 that describes how "AI is being adopted from the bottom-up, without clear top-down company-wide structures" and offers Lip a different perspective.

Organisational level

Define new Liip AI values, what the company stands for and against. Highlight what will not be replaced	Develop beginner-friendly pathways to address knowledge gaps between them and advanced AI users	Promote the value of human made work and ideas	Open pathways for designers to contribute to AI tool development	Develop a simple ethical checklist for designers to consider before using AI in client projects
Create guidelines for designers and developers to align their workflows and prevent unwanted overstepping	Collaborate with universities and research institutions to get new perspectives and insights	Avoid pushing designers to an "arbiter" role with no need	Customise HRM practices in accordance with employees' competencies	Promote transparency about AI use in projects
Develop a shared glossary of AI-human collaboration terms to facilitate communication between designers and developers	Create an internal platform with curated tools and prompts tailored for design projects and tasks	Adopt the Human Capability Standards (HCS) framework	Motivate designers with AI skills certifications	Incorporate Corporate Social Responsibility (CRS) models intro current model at Liip
Create a NO AI SLOP guideline to prevent uncritical use of AI	Make ready to use toolkits and guides about AI use, do's and dont's, use cases, etc.	Prioritise skills and competencies over role titles	Establish a rotating multidisciplinary AI revisory board	Create risk assessment analysis processes
Define efficiency-boosting tasks to be automated	Collaborate with stakeholders to access technology and knowledge	Establish R&D time for exploration protected in the schedule	Use tools with less ecological footprint	Work closely with the legal team to create legal frameworks
Communicate a narrative that prioritises collaboration and growth instead of just efficiency				

Design teams

Use the double diamond method as a guide to assign AI automated tasks, augmented tasks and NO AI tasks.	Create catalogue of previous and ongoing individual AI explorations to document what has been/can done, including prompts	Prioritise prevention and handling of AI hallucinations	Use AI to create a designer availability schedule to optimise project teams and protect designers time	Incentivise Liipers to be vocal about ethical issues
Start projects by allocating AI and human tasks	Let designers explore automation starting from mundane tasks	Pair less confident designers with more enthusiastic ones	Foster peer recognition for human-led creative decisions and contributions specifically	Promote critical thinking and analysis of AI outputs
Explore AI generated guides that define processes, define limits and adapt to each project	Use a collaborative, participatory approach to AI learning	Create creative AI challenges and display the results from the designers. Not as a competition but to showcase what can be done	Do not define how designers should work but offer support	Implement bias awareness training and processes of review
Apply human-in-the-loop framework and other practical AI-human collaboration methods	Foster inter-team workshops to build collective knowledge	Assign new tools like sketch to image or 3D generators to different designers to explore each quarter	Challenge creatives to use AI in unique ways and think bigger	Embed hallucination control processes in every instance of AI use

Individual designers

Adopt a NO AI SLOP principle	Explore tailored training and e-learning tools	Always start projects with human ideas that can be expanded with AI	Use AI as a powerful tool to combat design fixation
Define what tasks in the workflow are AI-eligible and which ones are not	Continue Liip talks where designers share what worked and what didn't	Reflect on and document what AI could not do on creative assignments	Use tools from less ethically ambiguous sources
Explore AI generated real-time coaching tailored for designers' needs	Prioritise human interaction during research and user testing, even if augmented with AI personas	Highlight the need for creativity, ingenuity and imagination in AI-augmented design projects	Focus on ethical decision-making and not just on outputs
			Be vocal about ethical issues

Figure 13 - Strategies assigned per layer

3.2.5.3 Strategic roadmap

To conclude, the roadmap (see figures 14 and 15) showcases an ideal timeline version of the AI adoption at Liip with the initiatives organised by resources needed. Compared to the previous methods, the roadmap made specific changes to the ideas on Phase 3 to reflect the evolution of the strategies across time. The timing, of course, will depend on Liip's availability and will have to adapt to the projects already being developed and future innovations that will be released. This roadmap serves as an initial version of a strategic timeline that should be approached as a loose guide and not a fixed structure.

The three phases span across one year, with the last one continuing into the future, and display initiatives that can be implemented immediately in the first couple of months, short-term measures that require resources and planning, and long-term strategies that will entirely depend on what worked during the first two stages.

Now	*Immediate*	3m	*Short term*	9m	*Long term*	12m+
Phase 1 Build the foundation		**Phase 2** Implement AI in practice		**Phase 3** Scale what works		
Establishing of principles and norms with no need for extra resources		**Active coordination and planning of AI infrastructure and processes**		**Systemic change through structural decisions and external partnerships**		
Create a NO AI SLOP guideline to prevent uncritical use of AI	Avoid pushing designers to an "arbiter" role with no need	Develop a shared glossary of AI-human collaboration terms to facilitate communication between designers and developers	Create creative AI challenges and display the results from the designers. Not as a competition but to showcase what can be done	Implement guides for automation and use cases as company policy	Use AI to create a designer availability schedule to optimise project teams and protect designers time
Communicate a narrative that prioritises collaboration and growth instead of just efficiency	Foster peer recognition for human-led creative decisions and contributions specifically	Define new Liip AI values, what the company stands for and against. Highlight what will not be replaced	Use AI as a powerful tool to combat design fixation	Formalise AI process that defines projects and limits	Prioritise skills and competencies over role titles
Start projects by allocating AI and human tasks	Promote the value of human made work and ideas	Explore AI generated guides that define processes, define limits and adapt to each project	Establish R&D time for exploration protected in the schedule	Collaborate with stakeholders to access technology and knowledge	Open pathways for designers to contribute to AI tool development

Apply human-in-the-loop framework and other practical AI-human collaboration methods	Prioritise human interaction during research and user testing, even if augmented with AI personas	Use the double diamond method as a guide to assign AI automated tasks, augmented tasks and NO AI tasks.	Assign new tools like sketch to image or 3D generators to different designers to explore each quarter	Collaborate with universities and research institutions to get new perspectives and insights	Customise HRM practices in accordance with employees' competencies
Let designers explore automation starting from mundane tasks	Do not define how designers work but offer support	Define efficiency-boosting tasks to be automated	Choose tools from less ethically ambiguous sources	Explore AI generated real-time coaching tailored for designers' needs	Establish a rotating multidisciplinary AI revisory board
Prioritise prevention and handling of AI hallucinations	Always start projects with human ideas that can be expanded with AI	Make ready to use toolkits and guides about AI use, do's and dont's, use cases, etc.	Embed hallucination control processes in every instance of AI use	Create an internal platform with curated tools and prompts tailored for design projects and tasks	Incorporate Corporate Social Responsibility (CRS) models intro current model at Liip
Foster inter-team workshops to build collective knowledge	Promote a critical reflection exercise after projects to document the creative limitations of AI	Create catalogue of previous and ongoing individual AI explorations to document what has been/can done, including prompts	Develop a simple ethical checklist for designers to consider before using AI in client projects	Adopt the Human Capability Standards (HCS) framework	Formalise tools with less ecological footprint
Continue Liip talks where designers share what worked and what didn't	Challenge creatives to use AI in unique ways and think bigger	Develop beginner-friendly pathways to address knowledge gaps between them and advanced AI users	Create risk assessment analysis processes		
Use a collaborative, participatory approach to AI learning	Incentivise Liipers to be vocal about ethical issues	Offer tailored training and e-learning tools	Work closely with the legal team to create legal frameworks		
Pair less confident designers with more enthusiastic ones	Promote critical thinking and analysis of AI outputs	Motivate designers with AI skills certifications	Implement bias awareness training and processes of review		

Highlight the need for creativity, ingenuity and imagination in AI-augmented design projects	Focus on ethical decision-making and not just on outputs
	Promote transparency about AI use in projects

Figure 14 - Strategies in a roadmap

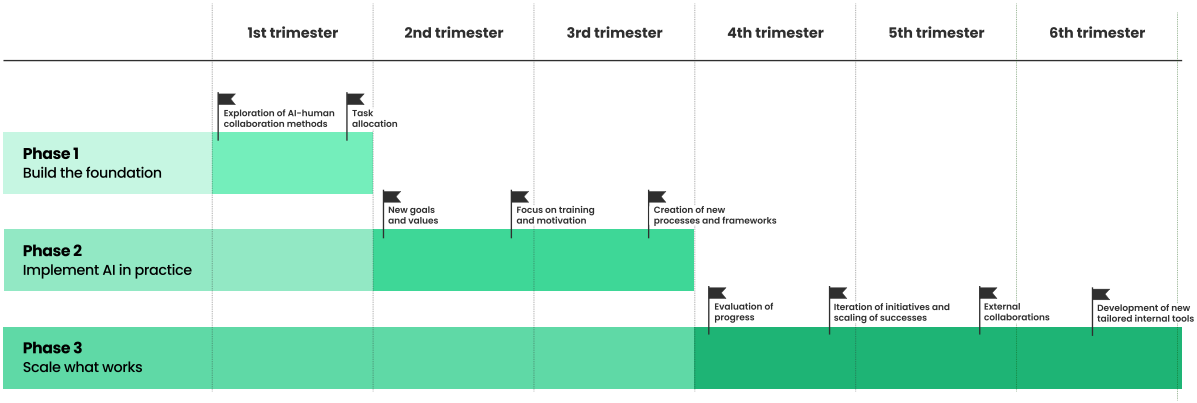


Figure 15 - Timeline of the roadmap with key moments

4. Discussion

4.1 Results

This research expands upon the existing theory that a successful AI adoption is not achieved by boosting productivity alone and that the process requires a greater focus on the psychological and social impact of the technology. The results suggest that this is a collaborative endeavour that needs the participation of leaders, teams, and individuals to work in parallel towards the same human-centred goals (Bowles & O'Hanlon, 2025). Ultimately, the use of the technology is not the primary interest, but how the company handles resistance and anxieties, reshapes its values, and redesigns the workflows to fit a new way of working and thinking.

AI adoption is a complex problem defined by a variety of domains, from financial and political, to social and philosophical. Often the main obstacle of the adoption process is the lack of a strategy that considers all of these different facets and takes into account both technological and human perspectives (Dwivedi et al., 2021). The data from Liip showcases this complexity,

where technical limitations are not actually the main barrier of AI transformation and that designers feel unprepared due to the perceived lack of conditions by the company and the industry as well as feeling psychological overwhelmed by the expectations put on them. Without the necessary systems to support a responsible AI-human collaboration, the human essence that defines the meaning and nuance of design artifacts is eroded for the sake of quantity. Consequently, creatives are put in a position where they feel encouraged to hand over their creative freedom and accept subpar results from the generative models (Garcia, 2025).

This position the study in contrast with some literature that locates the tension between human-AI collaboration at the level of tool use and workflows (Chui et al., 2018). As an example, Lee et al (2025) and Yu (2025) propose new processes with AI-augmentation throughout research, ideation, user testing and prototyping, that showcase the opportunities of the collaboration for creative outcomes. Although these contributions are valuable and well grounded, their implementation at Liip would assume a level of readiness in terms of collective knowledge, top-down support, and psychological safety that the study's empirical data suggests has not been achieved yet. Designers in this study were not just asking how to use AI better but wondering if it was safe to try at all. This adoption gap: the distance between AI's potential and designers' readiness to adopt it responsibly, is a necessary step to be addressed before introducing workflow interventions and must target motivation, confidence, professional growth, and ethical transparency. The importance of having a strong strategy and leadership commitment is highlighted by Sánchez et al. (2025) and Singla et al. (2025) as a prerequisite for a meaningful technological transformation. Although Liip's governance model does not follow a typical hierarchical leadership, the sentiment is still relevant as employees must align their values, define goals and to lead the company together. With this, the research does not directly contradict the existing literature but extends it into a more grounded organisational reality.

Ultimately, the resulting initiatives of this study put the psychological dimension in the centre of the strategic transformation. The idea that designer's resistance to adopt AI is a consequence of its threatening presence to creative intentionality and authorship (Garcia, 2025) was clearly reinforced when Liipers expressed their concerns that the creative core of their work will being handed to machines. While it might sound like a simply emotional issue, if we consider that Schön's (1992) depiction of designer know-how, explained as the tacit ability to see, judge and reflect on their conscious and subconscious moves, is a skill that machines fundamentally struggle to replicate, we can see how it feels as rational. In other words, machines are not yet capable of engaging in exploratory and transformational creativity and take over human-made

creative decisions with the level of contextual nuance and empathy needed (Boden, 2004). With emerging perspectives that frame designer as arbiters, who evaluate and direct AI outputs rather than producing them (Yu, 2025), it is necessary that designers continue to be engaged in the creation of design projects and feel like their authorship is protected. Like most creatives, designers enjoy the process of making things. Replacing this essential element from their work would not only go against their wants and needs, but it would also be a detriment to the industry in general.

4.2. Answering the Research Questions

Following the DRM method, this study answers the research questions through the different stages of the framework. The Descriptive Study I addresses the first sub-question regarding how designers currently collaborate with AI in their everyday work, which reveals a mostly individual practice with a big emotional toll. Many designers at Liip are already experimenting with AI augmentation across brainstorming and prototyping, AI-assisted coding, and supporting administrative tasks, but without a shared framework to guide their decisions. This mirrors Chanchamnan et al.'s (2022) observation that AI use in design tends to emerge from personal initiatives rather than organisational strategy, as well as Bowles and O'Hanlon's (2025) warning about bottom-up adoption without top-down structural support. However, the most significant insight is understanding how designers feel when using AI and not the fact that they use it. The tension between the enthusiasm for AI's possibilities and promises of efficiency, and the anxiety about its implications for their professional identity was the dominant experience across Liip's designers. This is consistent with the concept of technostress that emerges from an overwhelming and under-supported environment that traps designers in a cycle of indecision and fear (Reznik et al., 2026).

The second sub-question, about how processes can be adapted to support AI co-creation in organisations, is answered in the Prescriptive Study. Here, the results propose a way of redesigning processes that involves multiple active levels at the same time: establishing company-wide guidelines and values by the organisation, building collective knowledge at a team level, and fostering confidence and motivation for individual designers to engage with AI on their own terms. Across levels, co-design methods (Steen et al., 2011) emerge as the solutions for many barriers such as knowledge gaps and bottom-up implementation and support the idea that responsible technological transformation is a participatory process. Liip's Holacracy model,

which already distributes decision-making across employees, is particularly relevant to these approaches.

Together, the findings offer a grounded answer to the central research question. It is only possible for AI to function as a co-author in design processes when the conditions for a responsible adoption are in place. Before improving workflows through automation of repetitive tasks, making research and ideation more efficient, reducing cognitive load, and opening the possibilities for designers to explore other disciplines, the questions of human agency and ethics have to be analysed first. When designers feel genuinely supported and protected by the organisation they work in, they are more likely to engage with AI on their own terms, driven by curiosity rather than external pressure. That paradigm shift, from reluctant compliance to willing participation, is where responsible adoption actually begins. This means that building shared structures, creating safe environments for collaborative exploration, and continuously affirming the value of human creativity and ingenuity are the foundation of the technological transformation. Moreover, establishing ethical governance as a structural requirement instead of an afterthought is crucial (Madanchian & Taherdoost, 2025), just like maintaining active bias and data quality regulations (Chui et al., 2018). Bughin's (2025) warning about the gap between what organisations claim about responsible AI and what they actually practice is useful to reflect on here. The main risk for Liip is not having bad intentions but having insufficient structures to translate their good intentions into consistent action.

4.3 Limitations

As a case study, this research is both supported and constrained by its unique organisational context. Liip's Holacracy model, its culture, and its active AI exploration phase made it a useful environment for this kind of study. Designers were already engaging with the research questions, which resulted in richer data than what a more conservative organisation might have produced. At the same time, these characteristics mean the findings reflect a company that is already open to change, which is not the default condition for most companies. The same initiatives might not correspond with other organisations' challenges, goals, and resources, and the ideation process would look considerably different.

The strategies most applicable to other contexts are those in Phase 1 of the roadmap, the initiatives regarding building a strong foundation around guidelines, shared language, and psycho-

logical safety, precisely because they do not depend on a specific governance model to be feasible. The further along the roadmap, the more the strategies assume Liip's specific conditions, which require more adaptation from other organisations.

It is important to note that this research does not offer a concrete resolution to the central question it identifies. The strategies it proposes are a means to address the current issues in AI adoption but are in no way the final solutions to the tensions present. The gap between AI's transformative potential and designers' readiness remains an ongoing process at Liip, as it does in most organisations, with outcomes impossible to truly predict. The goal of this study was to offer a starting point to the transition and not a conclusion.

Additionally, there are several factors that limit the reliability of the conclusions that can be drawn from this study, particularly the data. Although obtained through a triangulation approach across observation, interviews, and a survey, the data pool is relatively small, with six structured interviews and a total of five survey responses from the study's primary target, designers. This means that the resulting analysis may reflect individual perspectives instead of a company-wide consensus. The use of paraphrasing of conversation notes rather than full transcripts also introduces a level of uncertainty in the qualitative data, as nuance may have been lost during the process. Finally, a testing and feedback phase would be necessary to further ground the strategies proposed, since without the ability to demonstrate them achieving the intended outcomes, they remain theoretical.

Overall, the evolving nature of AI also impacts the accuracy of the results, as some findings and strategies may become outdated sooner than expected, compared to research on more stable phenomena.

4.4 Future Research

Further research is needed to conclude the missing phases of the DRM framework, namely the Descriptive Study II and subsequent iterations, to refine the strategies through user testing, pilot projects, and feedback from a broader range of actors. A follow-up Descriptive Study would allow the strategies to be tested in practice and evaluated through measurable outcomes at Liip. This would facilitate the assessment of the initiatives on Phase 1 of the roadmap and whether they lead to producing the foundational conditions they aim to establish. Additionally, expanding the participant's pool to include other teams as well as collaborators and clients would offer

a bigger picture of the impact of AI adoption across the organisation rather than a singular discipline.

Most importantly, with AI technology continuously evolving, the research calls for ongoing revisiting of both findings and proposed strategies to ensure they maintain relevancy as the development of new tools, regulations, and industry norms can greatly affect their outcomes. Ideally, tracking the evolution of a single organisation through time and technological advancements would reveal necessary insights to understand if psychological and structural barriers lower over time thanks to controlled intervention or if new barriers emerge to replace them.

5. Conclusion

This research set out to explore how AI can function as a co-author in design practice without undermining the agency and creative vision of the designers who work alongside it. The findings suggest that answering this question requires a rethinking of how organisations support their people through transformational change rather than the implementation of new technologies.

The study's most significant contribution is the reframing of the problem, that AI adoption in design is not just a technical challenge. Resistance to adoption is shaped by fear, uncertainty, knowledge gaps, and the lack of structures that make responsible experimentation possible. Addressing these challenges demands a collaborative effort of leaders who set boundaries and strategic direction, teams who foster shared learning, and individual designers who are given the confidence and space to define how AI fits within their own creative workflows.

In this context, the transition towards an AI-supported workflow remains an ongoing process. Like many organisations navigating similar situations, the company’s experience demonstrates that AI integration is rarely as straightforward as it is often presented in existing literature. The strategies proposed in this study are therefore grounded in that reality, which is both their limitation and their strength.

Ultimately, this thesis argues that the future of design is defined by the continued shaping of its collaboration with AI, where designers draw the limits of what can be delegated to machines for the purpose of efficiency, and where human judgement, contextual awareness, creative direction, and empathy remain essential. Rather than diminishing the value of the design process,

AI challenges the profession to intentionally define and defend the uniquely human contributions that give meaning and quality to creative work.

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7. Appendices

Appendix A – Preliminary issue map

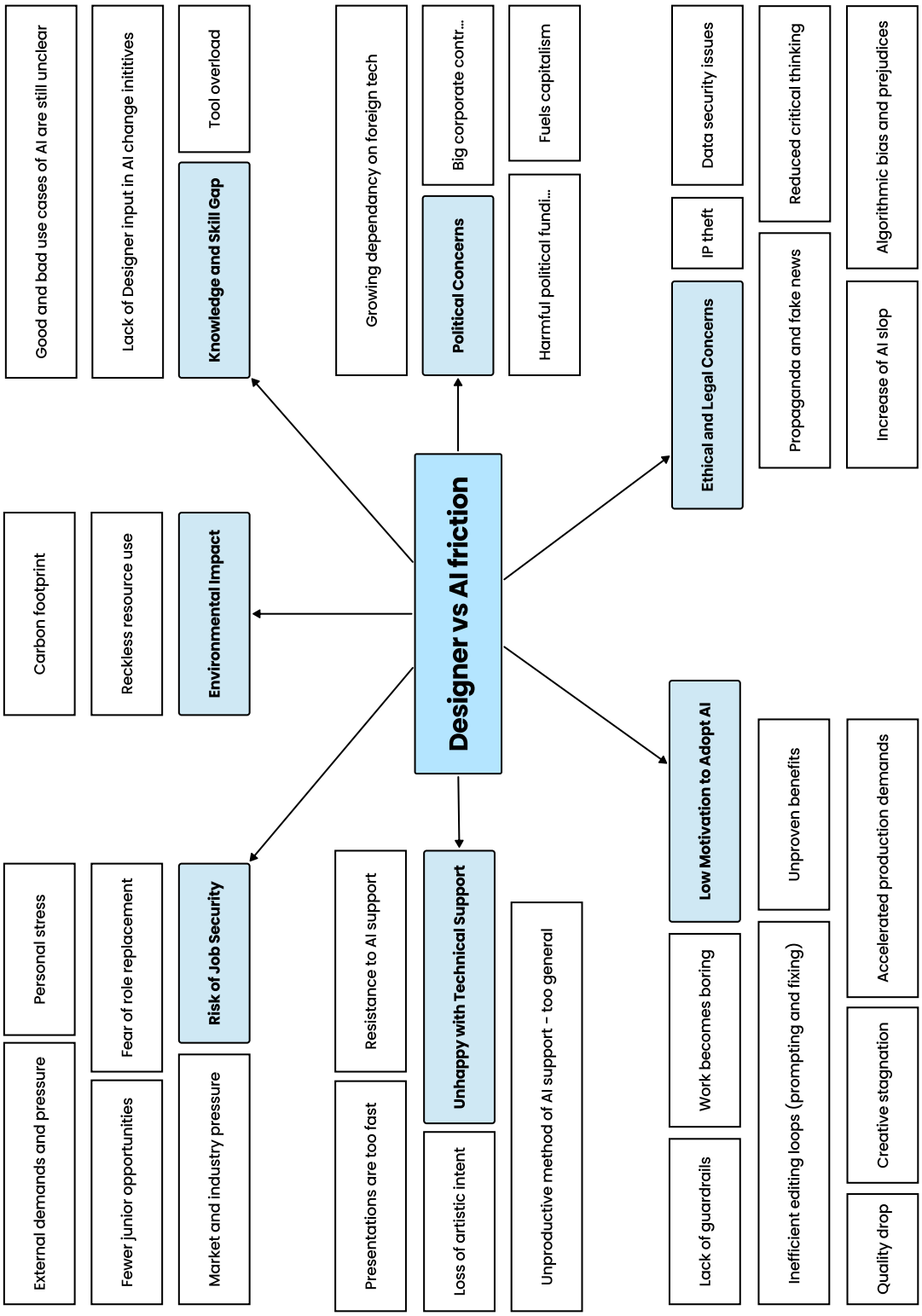


Figure A - Initial issue map based on survey answers

Appendix B – Thematic analysis

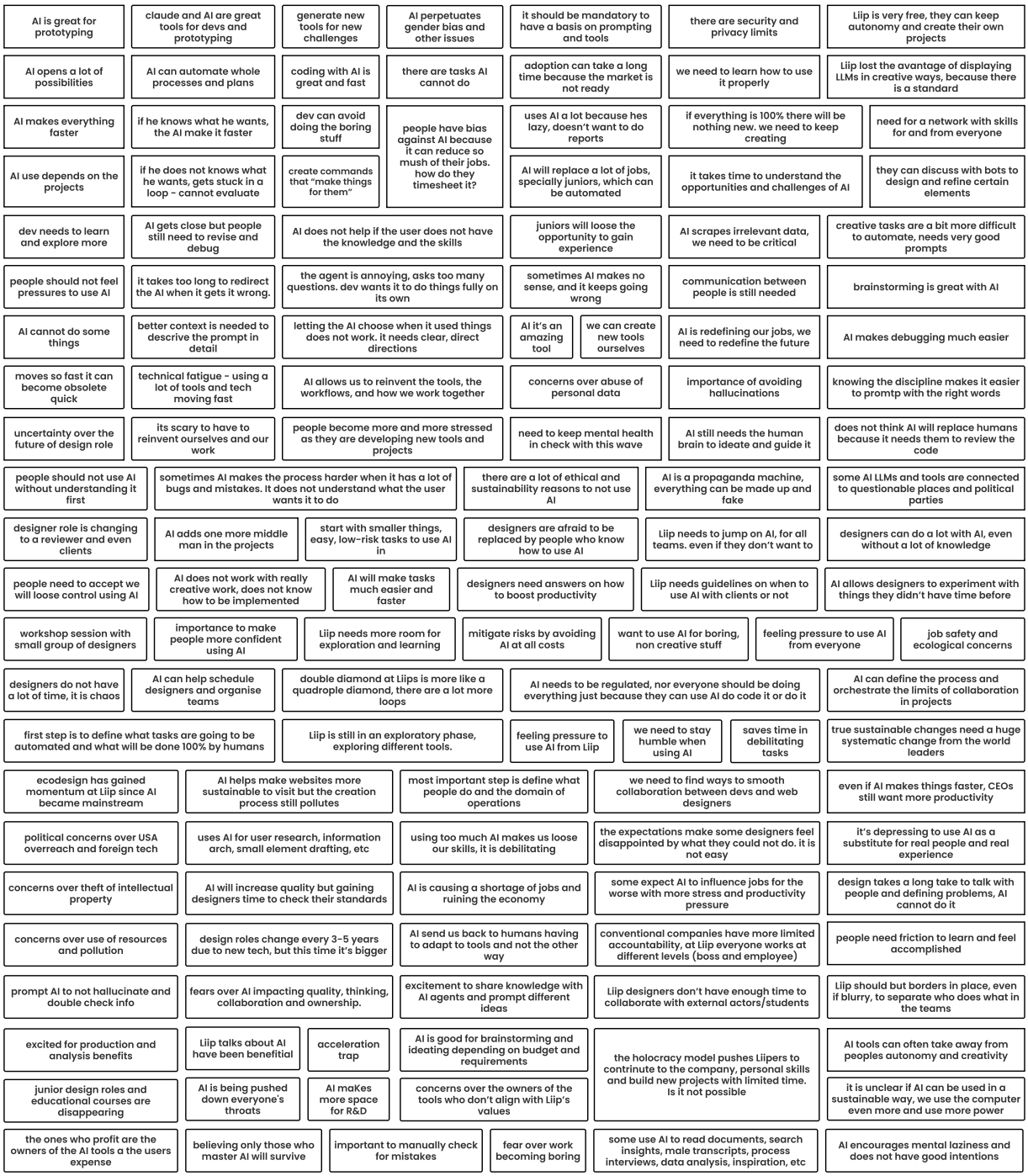


Figure B - Blocks of relevant information from notes and survey responses

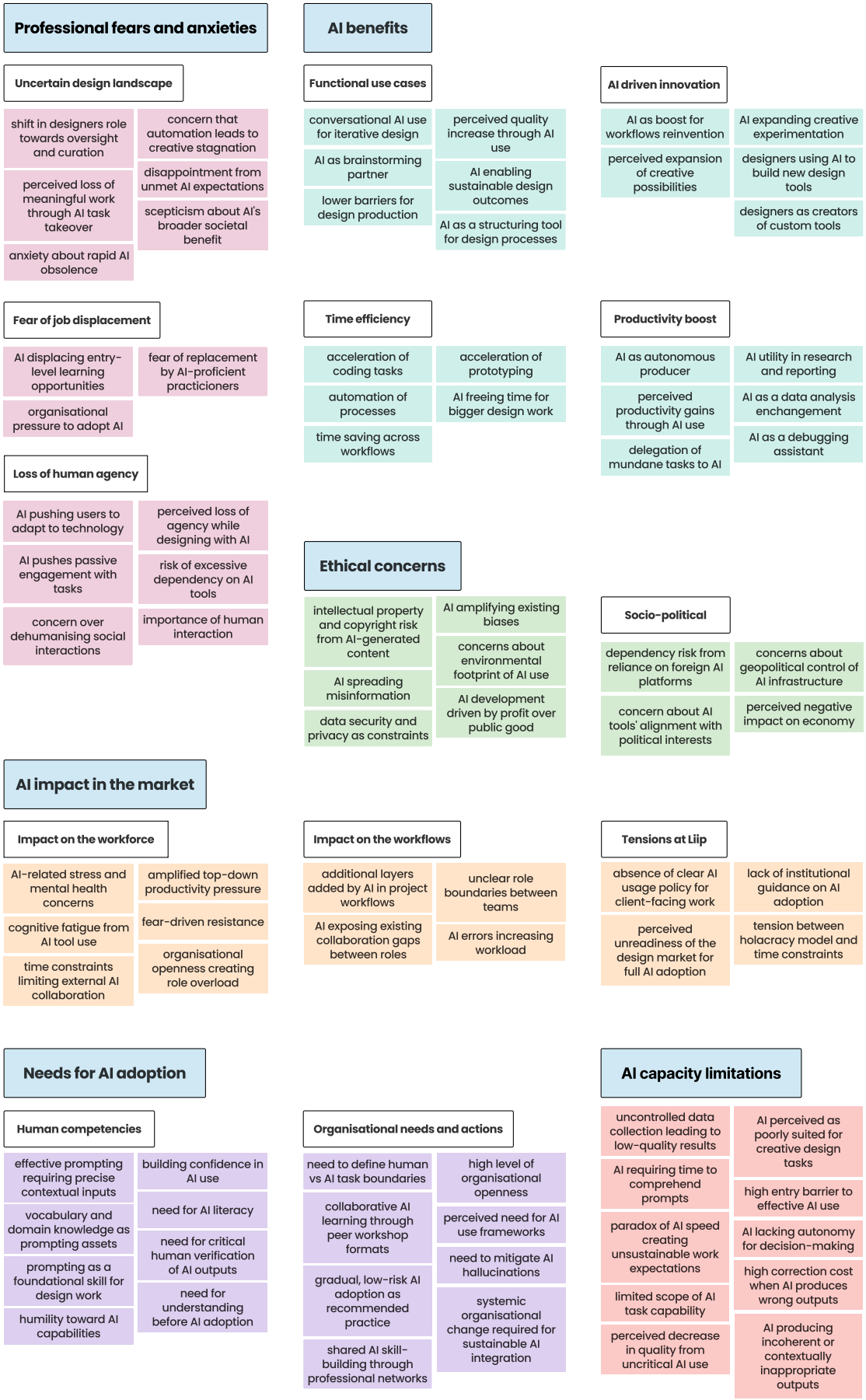


Figure C - Initial themes with corresponding codes

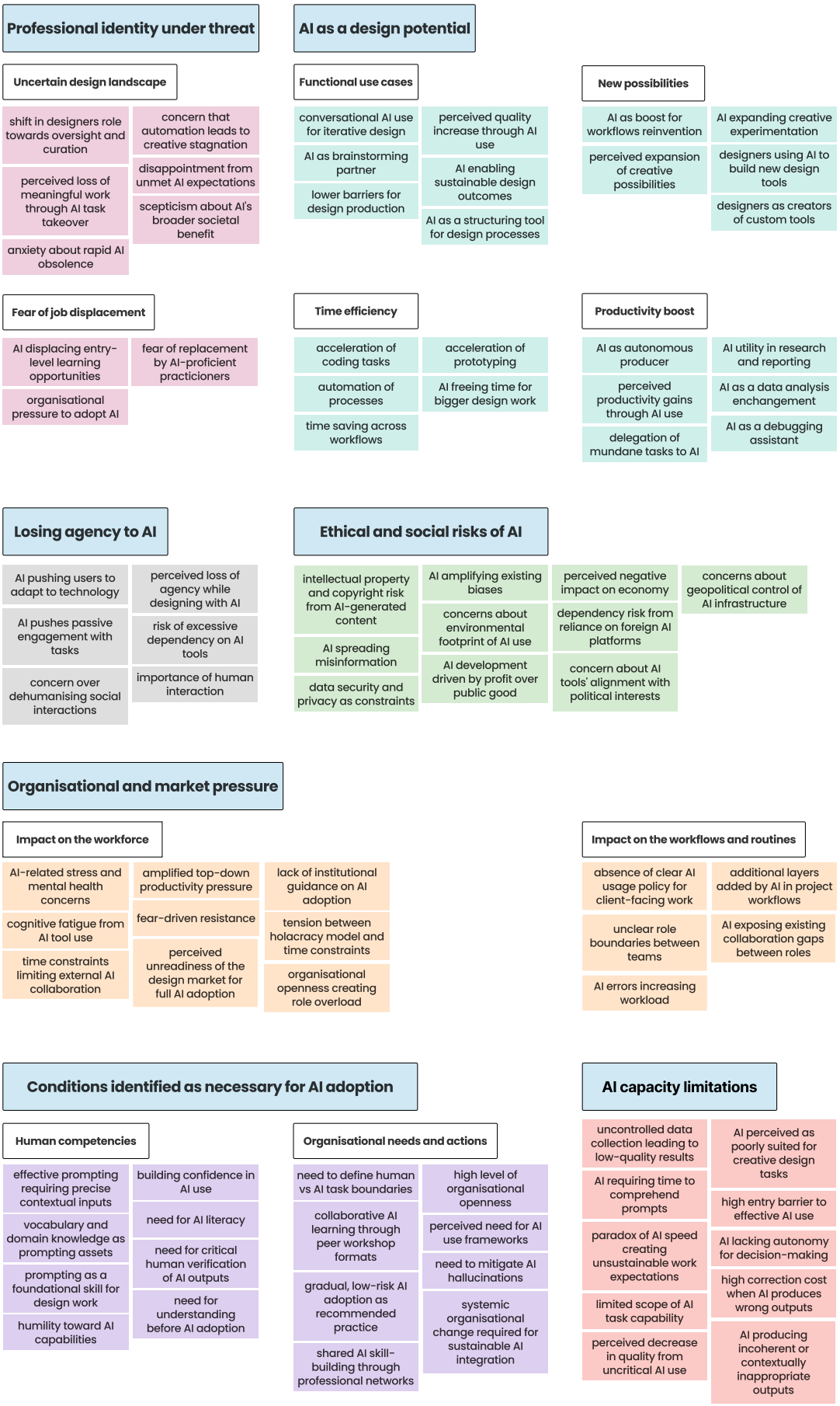


Figure D - Final themes with corresponding codes

Appendix C – Preliminary objectives tree

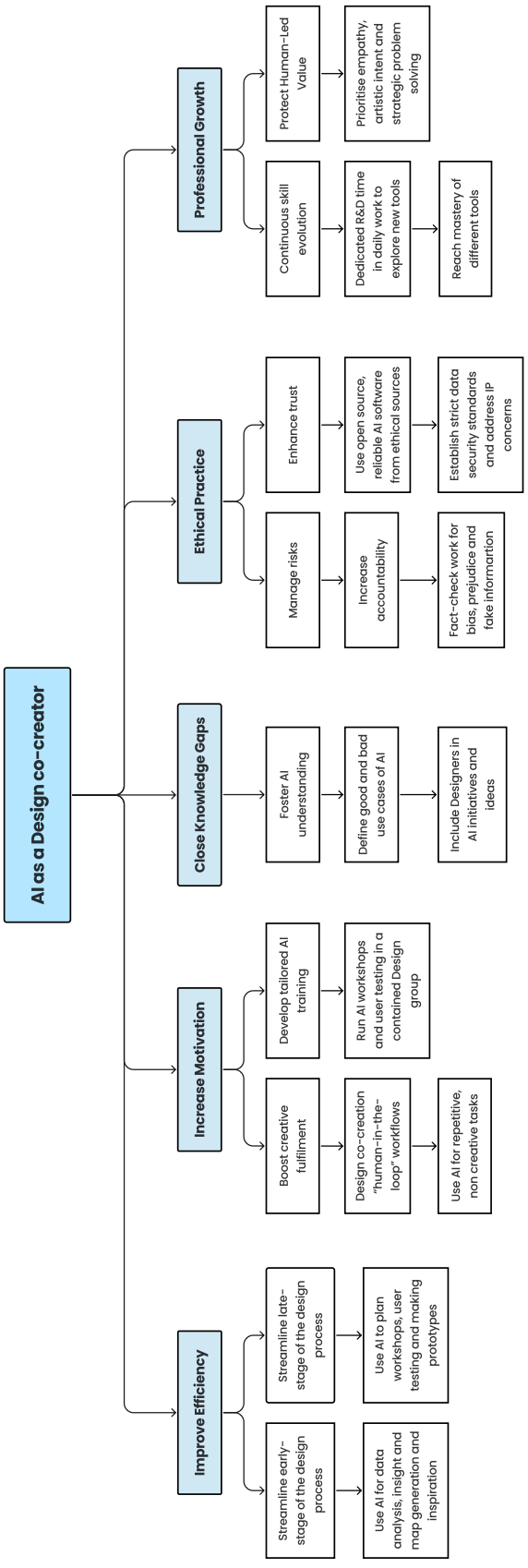


Figure E - Initial objectives tree based on preliminary issue map